An Introduction to Fractal Geometry, Illustrated by the Mandelbrot Set

New edition 2026

by

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Preface

This *Facharbeit*, "An Introduction to Fractal Geometry, Illustrated by the Mandelbrot Set", was prompted by an article in the German-language edition of "Scientific American". That publication presented the background to an object of recreational mathematics in an accessible form and showed fascinating computer graphics.

The aim of this Facharbeit is to clarify the mathematical framework behind such complex structures. The structures themselves conceal deep mysteries, and their investigation is constantly advancing.

It is obvious that a Facharbeit can treat the subject only to a limited extent, since this is a relatively new branch of discrete mathematics and the mathematical relationships are more complicated than they are often made out to be.

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Note on the new edition 2026

This edition was produced in 2026 on the basis of the 1990 Facharbeit. It follows the original's structure and line of argument, but has been reworded and considerably shortened: many calculations, examples, figures and individual sections of the original have been condensed or omitted, and mathematically incorrect statements have been corrected. Only some of these changes are marked individually in the highlighted note boxes and in the change log.

The 1990 original is preserved, together with a verbatim transcription and a facsimile of every page, in a private archive. The figures were recalculated or redrawn in 2026. The Python code used for this is documented in the appendix, separately from the original 1990 Pascal code.

Chapters 5 to 9 ("Further topics") were added in 2026. They build on chapters 1 to 4 and lead from the Mandelbrot set to Julia and Cantor sets, Hausdorff dimension, other complex functions and open questions in current research.

1 Introduction to complex numbers

1.1 General remarks

In algebra, when solving equations, for example when finding the zeros of a function, one frequently encounters the problem that an equation has no real solution. Within the real numbers, the square root of a negative number cannot be taken. This restriction can be overcome by extending the number system and introducing a new number i with the property $i^2 = -1$.

1.2 Definition of the complex numbers

A complex number consists of a real component and an imaginary component. The set of complex numbers is defined by

$$\mathbb{C} = \{\, a + bi \mid a,b \in \mathbb{R} \,\}, \qquad i^2 = -1.$$

For $z = a + bi$, $a = \mathrm{Re}(z)$ is called the real part and $b = \mathrm{Im}(z)$ the imaginary part. The notation $i = \sqrt{-1}$ is to be understood only as an intuitive shorthand; the actual definition is $i^2 = -1$.

1.3 The four basic arithmetic operations

Let $z_1 = a + bi$ and $z_2 = c + di$ with $a,b,c,d \in \mathbb{R}$.

1.3.1 Addition and subtraction

$$z_1 + z_2 = (a + c) + i(b + d), \qquad z_1 - z_2 = (a - c) + i(b - d).$$

Complex numbers are added or subtracted by adding or subtracting their real parts and their imaginary parts separately.

1.3.2 Multiplication

$$z_1 z_2 = (a + bi)(c + di) = (ac - bd) + i(ad + bc).$$

Here the brackets are multiplied out as with real expressions, and $i^2 = -1$ is then substituted. For the complex conjugate $\overline{z} = a - bi$ we have, in particular,

$$z\overline{z} = (a + bi)(a - bi) = a^2 + b^2.$$

1.3.3 Division

For $z_2 \neq 0$, that is $c^2 + d^2 \neq 0$, the numerator and denominator of the fraction are multiplied by the complex conjugate of the denominator:

$$\frac{z_1}{z_2} = \frac{a + bi}{c + di} = \frac{(a + bi)(c - di)}{(c + di)(c - di)} = \frac{ac + bd}{c^2 + d^2} + i\,\frac{bc - ad}{c^2 + d^2}.$$

Both components of the result are real numbers.

1.4 Basic laws of algebra in $\mathbb{C}$

The arithmetic laws familiar from the real numbers also hold for complex numbers. For $z_1,z_2,z_3 \in \mathbb{C}$ we have the commutative law

$$z_1 + z_2 = z_2 + z_1, \qquad z_1 z_2 = z_2 z_1,$$

the associative law

$$(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3), \qquad (z_1 z_2) z_3 = z_1 (z_2 z_3),$$

and the distributive law

$$z_1 (z_2 + z_3) = z_1 z_2 + z_1 z_3.$$

The identity elements are $0 = 0 + 0i$ for addition and $1 = 1 + 0i$ for multiplication. The additive inverse of $z = a + bi$ is $-z = -a - bi$. For $z \neq 0$ the multiplicative inverse also exists:

$$z^{-1} = \frac{\overline{z}}{z\overline{z}} = \frac{a - bi}{a^2 + b^2}.$$

The complex numbers with addition and multiplication therefore form a field.

> **Correction note** In the original, individual transformations were worked through in detail, but the identity elements and the multiplicative inverse were not stated in full. This gap, marked by the teacher, has been closed here.

1.5 The modulus (absolute value) in $\mathbb{C}$

The modulus of a complex number $z = a + bi$ is defined by

$$|z| = |a + bi| = \sqrt{a^2 + b^2}.$$

The following hold:

$$|z| \geq 0, \qquad |z| = |-z|, \qquad |z_1 z_2| = |z_1|\,|z_2|$$

and, for $z_2 \neq 0$,

$$\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}.$$

In addition, the triangle inequality holds:

$$|z_1 + z_2| \leq |z_1| + |z_2|.$$

To prove it, one can square both non-negative sides. With $z_1 = a + bi$ and $z_2 = c + di$ it then suffices to show

$$ac + bd \leq \sqrt{a^2 + b^2}\sqrt{c^2 + d^2},$$

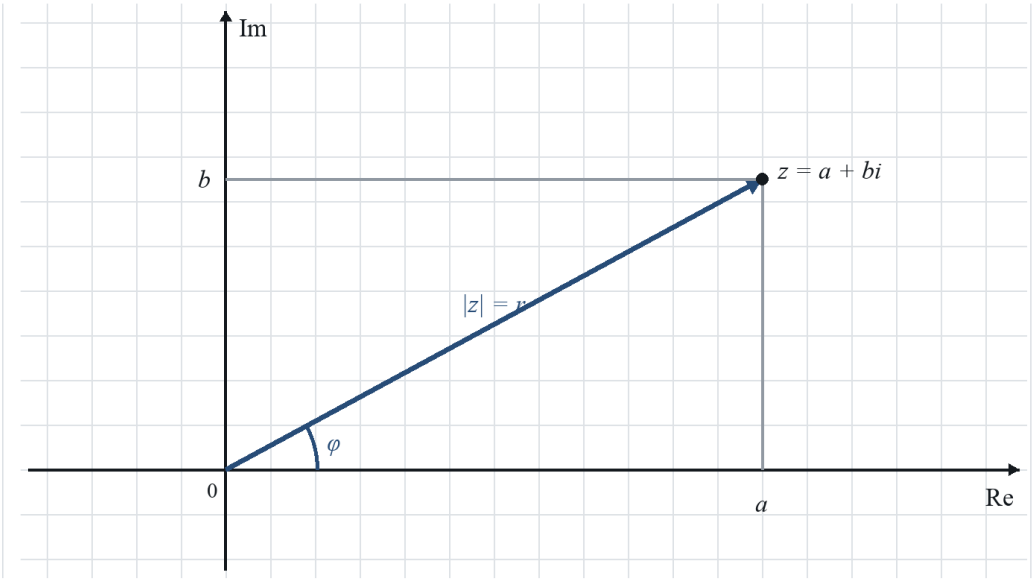
which is the Cauchy–Schwarz inequality in $\mathbb{R}^2$. Equivalently, the transformation used in the original ends with the true statement

$$0 \leq (ad - bc)^2.$$

In the complex plane (Gaussian plane), $z = a + bi$ corresponds to the point (a,b). The modulus $|z|$ is its distance from the origin, and $|z_1 - z_2|$ is the distance between the corresponding points. The conditions $|z| = 1$, $|z| < 1$ and $|z| > 1$ describe the unit circle, its interior and its exterior.

1.6 Geometric interpretation

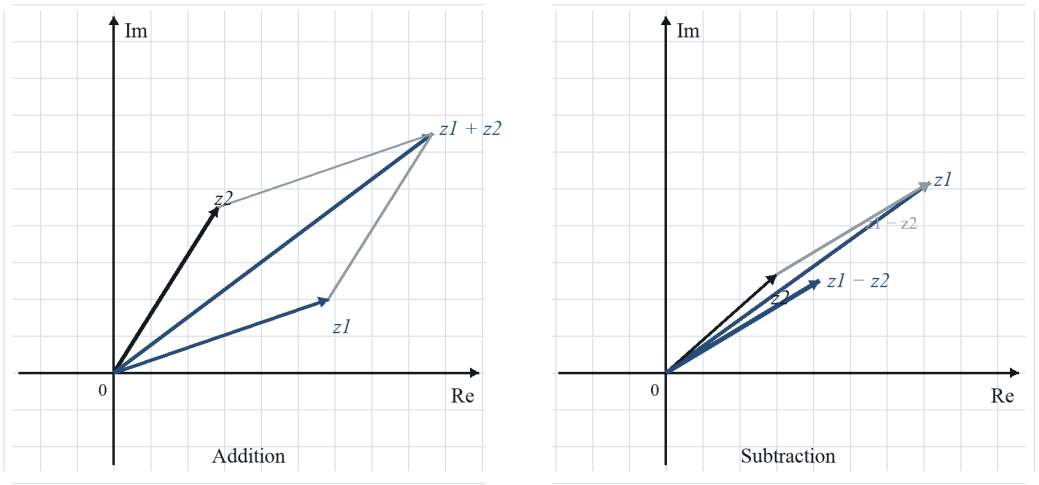
The real part is plotted on the horizontal axis of the complex plane and the imaginary part on the vertical axis. Addition and subtraction can thus be represented as vector addition and vector subtraction respectively.



The point $z = a + bi$ is determined by its two coordinates a and b. Equivalently, it can be described by its distance $r = |z|$ from the origin and the angle φ to the positive real axis.

1.6.1 Addition and subtraction

Every complex number can be regarded as a position vector. Completing the vectors to a parallelogram gives the sum of two complex numbers. The difference $z_1 - z_2$ is obtained by adding z_1 and $-z_2$.



When adding, $z_1 + z_2$ arises as the diagonal of the parallelogram. The difference $z_1 - z_2$ can be interpreted as the vector joining z_2 to z_1.

1.6.2 Multiplication and division

For $z \neq 0$ we introduce polar coordinates. With $r = |z| > 0$ and an argument φ we have

$$z = r(\cos\varphi + i\sin\varphi).$$

For

$$z_1 = r_1(\cos\varphi_1 + i\sin\varphi_1), \qquad z_2 = r_2(\cos\varphi_2 + i\sin\varphi_2)$$

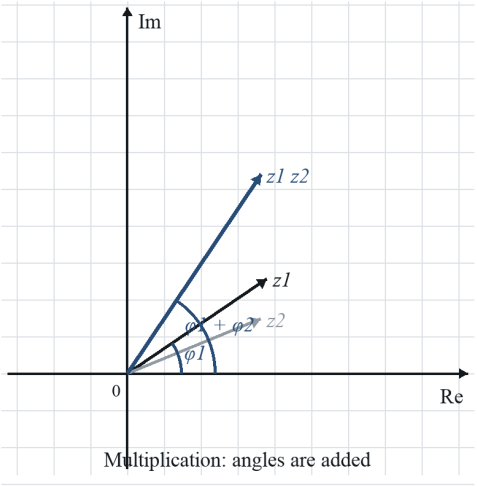
the angle-addition formulas give

$$z_1 z_2 = r_1 r_2\big(\cos(\varphi_1 + \varphi_2) + i\sin(\varphi_1 + \varphi_2)\big).$$

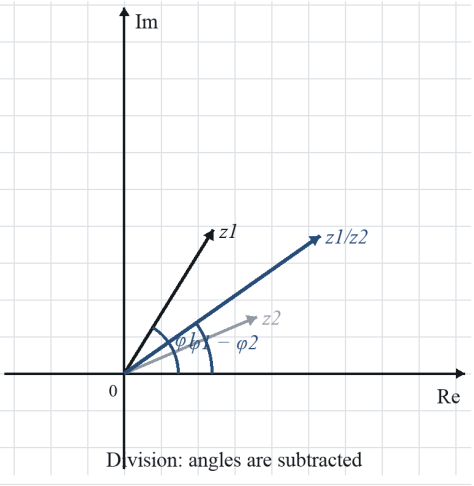
In multiplication, therefore, the moduli are multiplied and the arguments added. Correspondingly, for $z_2 \neq 0$,

$$\frac{z_1}{z_2} = \frac{r_1}{r_2}\big(\cos(\varphi_1 - \varphi_2) + i\sin(\varphi_1 - \varphi_2)\big).$$

In division, the moduli are divided and the arguments subtracted.



Multiplication: angles are added



Division: angles are subtracted

The drawings illustrate only the angle relationships. The length of the product vector or of the quotient vector also depends on the moduli $r_1 r_2$ or r_1/r_2 respectively.

1.7 Ordering of complex numbers

The complex numbers cannot be ordered in such a way that, together with the usual arithmetic operations, they form an ordered field. If $i > 0$, then compatibility with multiplication would give $i^2 = -1 > 0$. If, on the other hand, $i < 0$, then $-i > 0$, and likewise $(-i)^2 = -1 > 0$ would follow. Both contradict $1 > 0$.

As a mere set, $\mathbb{C}$ can of course be ordered, for instance lexicographically by real and imaginary part. Sorting by modulus and then by a fixed choice of argument is also possible. Such orderings, however, are not compatible with addition and multiplication in the way that the usual ordering of the real numbers is.

Correction note The original proof considered only the case $i > 0$, and the subsequent ordering by modulus and angle was not fully specified. Both of the teacher's remarks have been taken into account here.

1.8 Examples

Example 1 The equation $z^2 + 2 = 0$ has the two solutions

$$z_1 = i\sqrt{2}, \qquad z_2 = -i\sqrt{2}.$$

Example 2 For $z^4 + 3 = 0$, that is $z^4 = -3$, we write

$$-3 = 3\bigl(\cos(180^\circ) + i\sin(180^\circ)\bigr).$$

De Moivre's theorem gives the four solutions

$$z_k = \sqrt[4]{3}\left(\cos\frac{180^\circ + 360^\circ k}{4} + i\sin\frac{180^\circ + 360^\circ k}{4}\right), \qquad k = 0,1,2,3.$$

Their arguments are 45°, 135°, 225° and 315°.

Example 3 For $z^5 + 4 = 0$, that is $z^5 = -4$, we correspondingly have

$$z_k = \sqrt[5]{4}\left(\cos\frac{180^\circ + 360^\circ k}{5} + i\sin\frac{180^\circ + 360^\circ k}{5}\right), \qquad k = 0,1,2,3,4.$$

The five arguments are 36°, 108°, 180°, 252° and 324°. The value 342° given in the original was a slip of the pen.

2 Sequences and limits in $\mathbb{C}$

2.1 Sequences and convergence in $\mathbb{C}$

A complex sequence is a mapping

$$\mathbb{N} \longrightarrow \mathbb{C}, \qquad n \longmapsto z_n.$$

Writing $z_n = a_n + ib_n$, the real and imaginary parts each form a sequence of real numbers. A complex sequence is called convergent if the sequence of real parts and the sequence of imaginary parts both converge.

2.1.1 Definition of the limit

The sequence (z_n) converges to $z \in \mathbb{C}$ if for every $\varepsilon > 0$ there exists an $N \in \mathbb{N}$ such that

$$|z_n - z| < \varepsilon \qquad \text{for all } n \geq N.$$

One writes

$$\lim_{n\to\infty} z_n = z.$$

For $z_n = a_n + ib_n$ and $z = a + ib$,

$$z_n \longrightarrow z \quad \Longleftrightarrow \quad a_n \longrightarrow a \ \text{ and } \ b_n \longrightarrow b.$$

From $|z_n - z| < \varepsilon$ it follows immediately that $|a_n - a| < \varepsilon$ and $|b_n - b| < \varepsilon$. Conversely, one chooses the bound $\varepsilon/2$ for both real sequences; then

$$|z_n - z| \leq |a_n - a| + |b_n - b| < \varepsilon.$$

> **Correction note** In the original, only one direction of this equivalence was proved. The converse, which the teacher asked for, has been added. The definition of the limit must also be distinguished from the Cauchy criterion proper.

2.2 Limit theorems

Let $z_n \to z$ and $w_n \to w$. Then

$$z_n + w_n \longrightarrow z + w,$$
$$z_n - w_n \longrightarrow z - w,$$
$$z_n w_n \longrightarrow zw.$$

If $w \neq 0$, then w_n is also non-zero from some sufficiently large index onwards, and

$$\frac{z_n}{w_n} \longrightarrow \frac{z}{w}.$$

By the continuity of the modulus, it also follows that

$$|z_n| \longrightarrow |z|.$$

The statement about polar coordinates that followed in the original requires a restriction: if $z \neq 0$ and a continuous branch of the argument is chosen in a neighbourhood of z, then the correspondingly chosen arguments also converge. Without such a choice, angles are determined only up to multiples of 2π.

2.3 Example

Let

$$z_n = \frac{n^2 i^n}{n^3 + 1}.$$

Since $|i^n| = 1$,

$$|z_n| = \frac{n^2}{n^3 + 1} \leq \frac{1}{n} \longrightarrow 0.$$

Hence z_n converges to 0. For $\varepsilon > 0$ it suffices to choose some $N > 1/\varepsilon$, because then for all $n \geq N$

$$|z_n| < \varepsilon.$$

Geometrically, this means that from this index onwards all terms of the sequence lie inside the circle with centre 0 and radius ε.

> **Correction note** The original solution treated the numerator and denominator separately, even though their individual limits do not exist as finite numbers, and it reversed an ε-inequality incorrectly. The direct estimate of the modulus closes this gap without replacing the example.

3 The Mandelbrot set

3.1 General remarks

The Mandelbrot set, also known informally as the "Apfelmännchen" (literally "little apple man"), became widely known around 1980 through the work of Benoît B. Mandelbrot and is named after him. Mandelbrot worked at IBM's Thomas J. Watson Research Center in Yorktown Heights near New York. His book on the fractal geometry of nature made the new forms known beyond a small circle of specialists.

Mathematics increasingly concerned itself with complex feedback processes. Such processes have in principle long been known and, in simple cases, can be described by differential equations. Mandelbrot's work showed that iterated equations can also be used to investigate fragmented, irregular structures reminiscent of forms in nature, such as clouds or fern leaves. Computing the corresponding images became possible with fast computers.

An iteration uses the result of one computational step as the input of the next step. In general one writes

$$x_0 = k, \qquad x_{n+1} = f(x_n).$$

This gives

$$x_1 = f(k), \qquad x_2 = f(f(k)), \qquad x_3 = f(f(f(k))), \quad \ldots$$

The notation $f^2(x)$ here denotes the twofold application $f(f(x))$ and must not be confused with $(f(x))^2$. This procedure is called iteration or feedback.

3.2 Mathematical description of the Mandelbrot set

In this chapter we now attempt to present a complex function that opens up for us the world of aesthetic computer graphics. In principle, many complex functions can be iterated. For our example, the Apfelmännchen graphic, we choose the quadratic function

$$f_k(z) = z^2 + k, \qquad z = a + bi, \quad k = c + di, \quad a,b,c,d \in \mathbb{R}.$$

For the real and imaginary parts it follows that

$$f_k(a + bi) = (a^2 - b^2 + c) + i(2ab + d).$$

Every point of the complex plane is described by a complex number; its distance from the origin is the number's modulus. For a fixed constant k, the function is applied repeatedly, beginning with the starting value zero:

$$z_0 = 0, \qquad z_{n+1} = z_n^2 + k.$$

The first iterations are therefore

$$z_1 = k, \qquad z_2 = k^2 + k, \qquad z_3 = (k^2 + k)^2 + k, \quad \ldots$$

For the image construction that follows, the decisive question is whether this sequence remains bounded or eventually escapes to infinity.

Correction note Despite the starting value $z_0 = 0$, the original records the expression $k^2 + k$ as the first step. It is in fact the second term of the sequence, z_2. The calculation has therefore only been re-indexed; the original approach is retained.

Three cases for the special case $k = 0$

The following case distinction from the original is correct for the special mapping $f_0(z) = z^2$, that is, for $k = 0$. Throughout, the moduli of the complex numbers are meant.

a) If $|z| < 1$, then $|z^{2^n}| = |z|^{2^n} \to 0$. The sequence approaches the origin.

b) If $|z| = 1$, all terms of the sequence have modulus 1. The terms themselves, however, need not be equal.

c) If $|z| > 1$, then $|z|^{2^n}$ grows without bound; the sequence escapes to infinity.

For the Mandelbrot set, however, it is $f_k(z) = z^2 + k$ that is investigated. Here the summand k must not be omitted. Membership therefore cannot be decided by the three cases at the unit circle alone. A simple counterexample is $k = -1$:

$$0 \longrightarrow -1 \longrightarrow 0 \longrightarrow -1 \longrightarrow \cdots$$

The orbit remains bounded, although it does not converge to zero.

Teacher's remark The red question "Wieso entfällt das k?" ("Why does the k disappear?") identifies precisely the gap in the original transition. The case distinction is therefore not deleted but explicitly retained as the special case $k = 0$.

3.3 The "Apfelmännchen" algorithm

Sequences that tend towards a fixed value or towards a periodic sequence are attracted by an attractor. If, on the other hand, the moduli escape without bound, one can speak intuitively of the attractor infinity. For the Mandelbrot set, however, it is not required that the sequence converge to zero. What matters is whether the orbit starting at zero remains bounded:

$$\mathcal{M} = \left\{ k \in \mathbb{C} \;\middle|\; z_0 = 0, \quad z_{n+1} = z_n^2 + k, \quad (z_n) \text{ is bounded} \right\}.$$

Calculating individual sequences by hand can show their behaviour only for a few examples. To investigate a larger region of the complex plane, the repeated calculation is transferred to the computer. Essentially three loops are needed for this: two loops run through the pixels, and the third carries out the iteration for each pixel.

Correction note Only the incorrect membership criterion has been replaced. The original defined the Mandelbrot set via the convergence $f^n(z) \to 0$. The correct criterion is the boundedness of the critical orbit; the point under investigation is the parameter k, not an arbitrary starting value.

Carrying out the algorithm

As an example, we choose the region $-2 \le c \le 1$ and $-1.5 \le d \le 1.5$ of the complex plane. Each pixel is assigned a constant $k = c + di$. During the calculation, the variables a and b hold

the real and imaginary parts of the current term of the sequence.

1. From the image size and the chosen region, determine the step sizes for c and d.
2. Assign the parameter $k = c + di$ to the current pixel.
3. Set $a := 0$, $b := 0$ and the iteration counter $h := N$.
4. Compute
$$\mathrm{Re} := a^2 - b^2 + c, \qquad \mathrm{Im} := 2ab + d.$$
5. Set $a := \mathrm{Re}$, $b := \mathrm{Im}$ and $h := h - 1$.
6. Repeat the calculation until $a^2 + b^2 > 4$ or $h = 0$.
7. If $h = 0$, the pixel is marked as not having escaped within the iteration limit. Otherwise it is certainly not a point of the Mandelbrot set.

The image region produced in this way is an approximation. With a finite number of iterations, a coloured point does not mean that membership has been proved, but only that the sequence has not escaped up to that point.

> **Correction note** In the original scheme the real and imaginary parts are computed but not assigned back to the variables a and b. Without this update no genuine iteration takes place. The two assignments have been added; the surviving Pascal listing also performs them.

Escape radius and Julia sets

Depending on the computing speed, the maximum number of iterations can be chosen, for example, between 25 and 150. A larger number improves the approximation, especially at the boundary, but increases the computing time.

For the critical orbit, the radius 2 suffices as the escape radius. If $|k| > 2$, then $z_1 = k$ already lies outside this circle. For $|k| \le 2$ and a later orbit point with $|z_n| > 2$ it follows that

$$|z_{n+1}| = |z_n^2 + k| \ge |z_n|^2 - |k| \ge |z_n|^2 - 2 > |z_n|.$$

The modulus then continues to grow and the orbit escapes to infinity. The equality $|z_n| = 2$ on its own, however, is not a uniform special case. For $k = -2$ the critical orbit reaches the value 2 and stays there; no general statement follows from this for all parameters or orbit points with modulus 2.

For comparison, we now consider the Julia sets. Mandelbrot and Julia sets arise from the same family of functions, but through two different questions:

- For the Mandelbrot set, the parameter k is varied while the starting value $z_0 = 0$ remains fixed.
- For a filled Julia set K_k, k is held fixed and the starting point z_0 is varied.

The two are linked by the important relationship

$$k \in \mathcal{M} \quad \Longleftrightarrow \quad K_k \text{ is connected.}$$

Here “connected” does not mean that any two points can necessarily be joined by a polygonal path within the set. That property would be stronger than the connectedness required here.

> **Historical context** The separation of the parameter plane and the dynamical plane, and the relationship between $k \in \mathcal{M}$ and the connectedness of K_k, were part of the theory in 1990. They are historical corrections and not modern standards applied after the fact.

3.4 Different ways of depicting Mandelbrot sets

Mandelbrot sets can be depicted in many ways. Depending on the chosen configuration, for instance a colour or a black-and-white display, different effects can be achieved. If the iteration process is stopped as soon as the modulus exceeds the given escape radius, the number of iterations needed up to that point can be assigned a colour or a grey level. This yields, among others, the following possibilities:

a) Only those points whose orbit does not escape within the iteration limit are marked.

b) Points with the same escape time receive the same colour. This produces coloured bands reminiscent of contour lines.

c) Combining the two methods gives a differentiated depiction with many gradations.

d) The escape speed can be interpreted as height. This produces a depiction that appears three-dimensional, for example a “Mandelbrot lake” with surrounding plateaus.

e) Assigning a different colour to each iteration step produces particularly colourful images.

f) Replacing colours with tones allows a musical composition to be created from the iteration data.

Many other forms of depiction, and variations on them, are also possible.

3.5 Similar structures within the Mandelbrot set

A further aspect directly connected with the Mandelbrot set is its approximate self-similarity or scale invariance. Objects are called self-similar if, when magnified, they show similar structures at different scales.

If one examines small regions at the boundary of the Mandelbrot set and magnifies them, ever new details appear. Miniatures of the Apfelmännchen shape occur that are related to the main figure. The Mandelbrot set, however, is not exactly self-similar everywhere; its repetitions are distorted and surrounded by new structures.

Such self-similar or approximately self-similar patterns are also found in nature, for instance in the branching of trees, ferns or river systems.

3.6 Further findings about Mandelbrot sets

The quadratic function investigated in this paper is the simplest case. Polynomials of higher degree can also be iterated and produce their own parameter and dynamical sets. For polynomials

of degree three, several critical points must in general be taken into account. Additional complex parameters can lead to higher-dimensional parameter spaces; a single complex variable, by contrast, remains representable in the two-dimensional complex plane.

Many depictions possess symmetries. Other structures show a sensitive transition between order and chaotic behaviour. Chaos theory investigates how very complicated behaviour can arise from simple deterministic rules.

Fractal geometry is not only a subject of theoretical research. Its basic ideas can also be conveyed in teaching and education, provided that a distinction is drawn between illustrative images, numerical approximations and mathematically proven statements.

4 Fractal dimension

4.1 The Koch snowflake

Classical Euclidean geometry essentially describes smooth shapes such as lines, circles, rectangles, spheres and pyramids. Many shapes in nature, however, are more complicated. A well-known example of a fractal curve is the Koch snowflake.

One starts with a line segment as the initiator. This is divided into three parts of equal length; the middle third is replaced by two sides of an equilateral triangle. The resulting polygonal line is called the generator. If each of its four segments is again replaced according to the same rule, and this process is continued indefinitely, the result is the Koch curve.

In the process the boundary becomes ever longer, while the resulting figure remains within a bounded region. The curve does not have the ordinary integer dimension of a line. It is more intricately branched than a line, yet it does not fill an area. For such fractured shapes Benoît Mandelbrot introduced the term “fractal”.

4.2 Fractal dimension

Suppose that each stage of an exactly self-similar figure consists of N reduced copies of the whole, each copy being scaled down by the factor $s > 1$. Then

$$N = s^D \qquad \Longrightarrow \qquad D = \frac{\log N}{\log s}.$$

This number D is called the similarity dimension. For a line segment, a square and a cube with the scaling factor $s = 3$, one obtains $N = 3$, $N = 9$ and $N = 27$, and hence the dimensions 1, 2 and 3.

For the Koch curve, each segment gives rise to four new segments, each scaled down by the factor 3. Therefore

$$D = \frac{\log 4}{\log 3} \approx 1.26186.$$

The dimension is thus not an integer, but lies between the dimension of a line and the dimension of a surface.

For sets that are not exactly self-similar, the formula above cannot be applied directly. A common numerical approximation is the box-counting method: one covers the set with squares of side length ε, counts the number $N(\varepsilon)$ of squares needed and examines the ratio

$$\frac{\log N(\varepsilon)}{\log(1/\varepsilon)} \quad \text{as} \quad \varepsilon \to 0.$$

> **Historical context** The original left it unclear whether the dimension referred to the whole Mandelbrot set or to its boundary. The Mandelbrot set itself has interior points and therefore has Hausdorff dimension 2. That its boundary also has Hausdorff dimension 2 was proved only after 1990 and must not be treated retrospectively as having been known at the time.

Further topics (2026)

5 Julia sets and connectedness

What you already know

- Complex numbers can be represented as points of the plane; the modulus $|z|$ is their distance from the origin (Chapter 1, Section 1.5).
- A sequence is called bounded if all its terms lie in a fixed circle about the origin (Chapter 2).
- For the Mandelbrot set one iterates $z_{n+1} = z_n^2 + k$ with the fixed starting value $z_0 = 0$ and asks for which parameters k the orbit remains bounded (Sections 3.2 and 3.3).
- As soon as a point of the orbit lies outside the circle of radius 2, the critical orbit is certain to escape (Section 3.3).
- Shades of grey or colours can show after how many steps a point escapes (Section 3.4).
- At the end of Section 3.3 the Julia sets were already mentioned briefly. Here we treat them in detail.

In Chapter 3 a single question stood at the centre: for which parameters does the orbit of the starting value 0 remain bounded? This chapter turns the question around. We hold the parameter fixed and let the starting value move. In this way each parameter gives rise to its own image, the Julia set. In the end it will turn out that the Mandelbrot set is something like a table of contents of all these images.

On notation. In Chapter 3 the parameter was called $k = c + di$; there c and d were real numbers. From now on we follow the specialist literature and call the complex parameter itself c. We therefore write

$$f_c(z) = z^2 + c, \qquad c \in \mathbb{C},$$

and denote the orbit of a starting value z_0 by z_0, $z_1 = f_c(z_0)$, $z_2 = f_c(z_1)$, $\ldots$

5.1 One fixed parameter, many starting values

We first choose the simplest parameter $c = 0$, that is, $f_0(z) = z^2$. The three cases from Section 3.2 answer the question completely: starting values with $|z_0| < 1$ run into the origin, starting values with $|z_0| > 1$ escape, and on the unit circle the modulus always remains 1. The set of all starting values with a bounded orbit is therefore the closed disc of radius 1; its boundary is the unit circle.

It is exactly this point of view that we generalise. For a fixed parameter c,

$$K_c = \left\{ z_0 \in \mathbb{C} \;\middle|\; \text{the orbit } z_0, z_1, z_2, \ldots \text{ under } f_c \text{ is bounded} \right\}$$

is called the *filled Julia set* of f_c. Its boundary

$$J_c = \partial K_c$$

is called the *Julia set*. Thus J_c consists of the points every neighbourhood of which, however small, contains both starting values with a bounded orbit and escaping starting values [10]. For $c = 0$, K_0 is the disc and J_0 the unit circle.

All remaining points form the *Fatou set*. For our quadratic functions it consists of two parts: the interior of K_c and the set of escaping starting values. Intuitively, the Fatou set is the "calm" part of the plane: if one shifts the starting value there a little, the orbit also behaves only a little differently. On the Julia set, by contrast, a tiny change of the starting value can decide whether the orbit remains bounded or escapes. In technical language one says that the iterates form a *normal family* on the Fatou set [10].

An escape criterion for arbitrary starting values

In Section 3.3 the radius 2 sufficed, because there one always started at $z_0 = 0$. With arbitrary starting values one has to be somewhat more careful. The following holds: if

$$|z| > \max\bigl(2,|c|\bigr),$$

then the orbit of z escapes. The justification is the same triangle inequality as in Chapter 3: since $|c| < |z|$, it follows that

$$|z^2 + c| \;\geq\; |z|^2 - |c| \;>\; |z|^2 - |z| \;=\; |z|\bigl(|z| - 1\bigr).$$

Since $|z| - 1 > 1$, we have $|z^2 + c| > |z|$; the new point of the orbit therefore satisfies the condition again. Hence the modulus grows in every step by at least the fixed factor $|z| - 1 > 1$ (with the starting value z), and the orbit escapes. In particular, K_c always lies in the disc of radius $\max(2,|c|)$; for the images in this chapter a bounded region of the plane is therefore sufficient.

Some simple properties

Because $f_c(-z) = f_c(z)$, z and $-z$ have the same orbit from the first step onwards. Therefore every filled Julia set is point-symmetric about the origin. If one replaces c by the conjugate number $\bar{c}$, the whole image is reflected in the real axis; for real c, K_c is therefore also mirror-symmetric about the real axis and, together with the point symmetry, about the imaginary axis.

Moreover: if z lies in K_c, then $f_c(z)$ and the two preimages $\pm\sqrt{z - c}$ also lie in K_c, since apart from one step they have the same orbit. K_c is therefore called *completely invariant*; the same holds for its boundary J_c [10]. Because f_c maps small pieces of J_c onto larger pieces, and the preimage branches map large pieces back onto small ones, shapes repeat on an ever smaller scale, albeit slightly distorted. The Julia sets are therefore, like the Mandelbrot set in Section 3.5, only approximately self-similar.

> **Historical context** The theory of iterated rational functions was founded by Gaston Julia [11] and Pierre Fatou [12, 13, 14] independently of each other. Julia's memoir of 1918 was awarded the Grand Prix by the Paris Académie des Sciences. There were no computer images at the time; the sets were known only through their properties. The names "Julia set" and "Fatou set" became customary only later. In 1980 Mandelbrot published one of the first computer graphics of the set named after him [15]; the name "Mandelbrot set" was coined by Douady and Hubbard [16].

5.2 A gallery

An image of K_c is produced by the same algorithm as in Section 3.3, only with the roles exchanged: each pixel is now a starting value z_0, and the parameter c stays the same for the

whole image. Figure 1 shows eight examples.

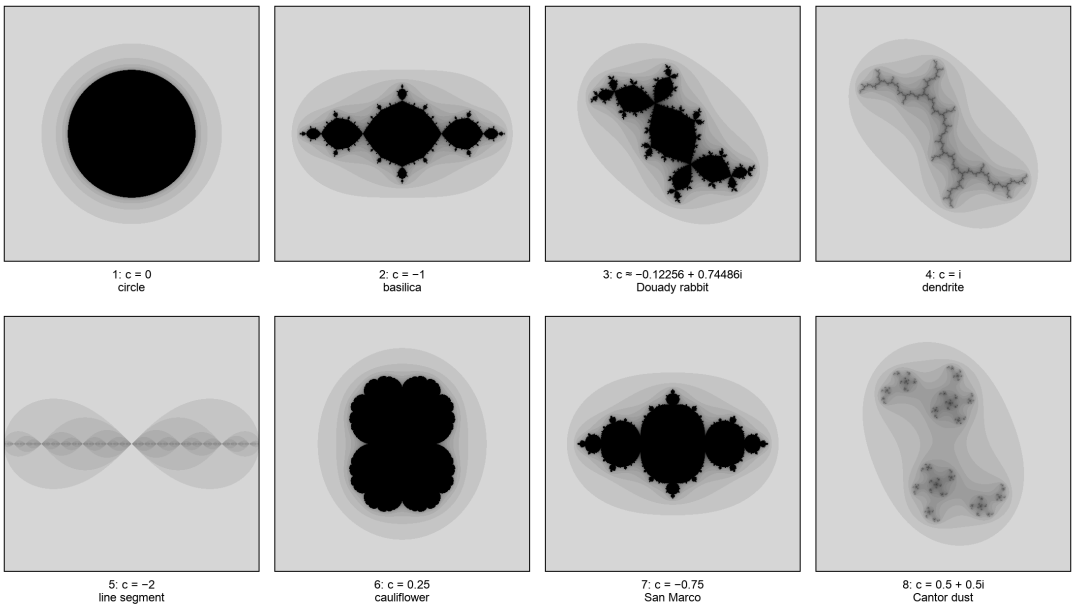


Figure 1: Eight filled Julia sets K_c in the dynamical plane (z-plane). Black: starting values whose orbit does not escape within the iteration limit. Shades of grey: escape time, light for fast, dark for slow escape. Images 1 to 7 belong to parameters in the Mandelbrot set and are connected; image 8 belongs to a parameter outside it and falls apart into dust.

1. $c = 0$, *circle*: K_0 is the disc, J_0 the unit circle.
2. $c = -1$, *basilica*: the orbit of 0 jumps back and forth between 0 and -1 (Section 3.2). K_{-1} consists of infinitely many black "bubbles" that touch at single points.
3. $c \approx -0.12256 + 0.74486i$, *Douady rabbit*: here the orbit of 0 returns exactly to 0 after three steps. Bubbles meet three at a time in a point.
4. $c = i$, *dendrite*: the orbit of 0 is $0 \to i \to -1 + i \to -i \to -1 + i \to \cdots$; after a preperiod it thus becomes periodic. K_i has no interior; $K_i = J_i$. The set looks like a finely branched tree.
5. $c = -2$, *line segment*: $K_{-2} = J_{-2}$ is the segment $[-2, 2]$ on the real axis.
6. $c = 0.25$, *cauliflower*: the orbit of 0 approaches the fixed point $\frac{1}{2}$ only very slowly.
7. $c = -0.75$, *San Marco*: Mandelbrot gave this set its name. It is a limiting form of the basilica.
8. $c = 0.5 + 0.5i$, *Cantor dust*: the orbit of 0 escapes after a few steps. K_c consists of uncountably many points, no two of which are connected to each other.

The names and descriptions follow [4, 10]. Even the first glance reveals two kinds of images: the first seven each hang together "in one piece", while the eighth has crumbled into dust. This observation will become the most important result of the chapter.

5.3 What "connected" means

Intuitively, a set is *connected* if it consists of one piece. Mathematically one formulates this the other way round: a set A is called connected if it cannot be split into two non-empty parts that are "separated" from each other, that is, held apart by two disjoint open sets.

A stronger property is *path-connectedness*: any two points of A can be joined by a continuous curve that runs entirely within A. Every path-connected set is connected, but not conversely. A well-known counterexample is the graph of $\sin(1/x)$ for $0 < x \leq 1$ together with the segment from $(0, -1)$ to $(0,1)$: the set is connected, yet no curve within it leads from the graph to the segment. As already noted in Section 3.3, what is meant in the following is always the weaker notion "connected".

5.4 The key: the orbit of the origin

Why does the starting value 0 in particular play a special role? The derivative $f_c'(z) = 2z$ vanishes only at $z = 0$. One therefore calls 0 the *critical point* of f_c, and its orbit the *critical orbit*. The Mandelbrot set is exactly the set of parameters for which the critical orbit remains bounded. In the new language this reads:

$$c \in \mathcal{M} \quad \Longleftrightarrow \quad 0 \in K_c .$$

The critical point now decides the shape of the whole set K_c. For the quadratic functions the following holds [12, 11, 16, 10]:

$$\boxed{c \in \mathcal{M} \quad \Longleftrightarrow \quad 0 \in K_c \quad \Longleftrightarrow \quad K_c \text{ is connected}}$$

The same holds for J_c. This is why $\mathcal{M}$ is also called the *connectedness locus* of the quadratic family. If c lies outside $\mathcal{M}$, then K_c even falls apart completely into dust. This dichotomy, its justification and the notion of the Cantor set are the subject of Chapter 6.

The idea of the proof in one sentence: as long as the critical point does not escape, the regions that approximate K_c from outside each remain a single piece; if it escapes, they are torn apart further and further. This is worked out in Chapter 6.

One small consequence is particularly vivid. Because $c = f_c(0)$, the orbit of c simply begins one step later than the orbit of 0. The one is bounded if and only if the other is. Hence we also have

$$c \in \mathcal{M} \quad \Longleftrightarrow \quad c \in K_c .$$

The parameter lies in "its" filled Julia set if and only if it belongs to the Mandelbrot set.

5.5 Two planes: dynamical plane and parameter plane

Because both images look black and grey and both kinds of set are drawn in the complex plane, an idea suggests itself that one occasionally reads in popular accounts: "The Julia set is a part of the Mandelbrot set." This statement is *false*, and it is worth understanding why.

The two sets live in different planes. K_c and J_c consist of *starting values* z_0; they lie in the *dynamical plane* (z-plane), and for each c there is a separate such plane. $\mathcal{M}$ consists of *parameters* c; it lies in the *parameter plane* (c-plane), and there is only one of those. Both planes

are copies of $\mathbb{C}$, so the images can be laid on top of each other. The same number, however, means a starting value in the one plane and a parameter in the other, that is, something entirely different.

If one nevertheless takes the statement literally, it is already false for simple examples [10]:

- For $c = 0$, J_0 is the unit circle, so $z = 1$ lies on J_0. As a parameter, however, 1 does not belong to $\mathcal{M}$: the orbit $0 \to 1 \to 2 \to 5 \to 26 \to \cdots$ escapes.
- For $c = -2$, $J_{-2} = [-2, 2]$. On the real axis, however, $\mathcal{M}$ extends only from -2 to $\frac{1}{4}$. The points between $\frac{1}{4}$ and 2 lie in J_{-2} but not in $\mathcal{M}$.

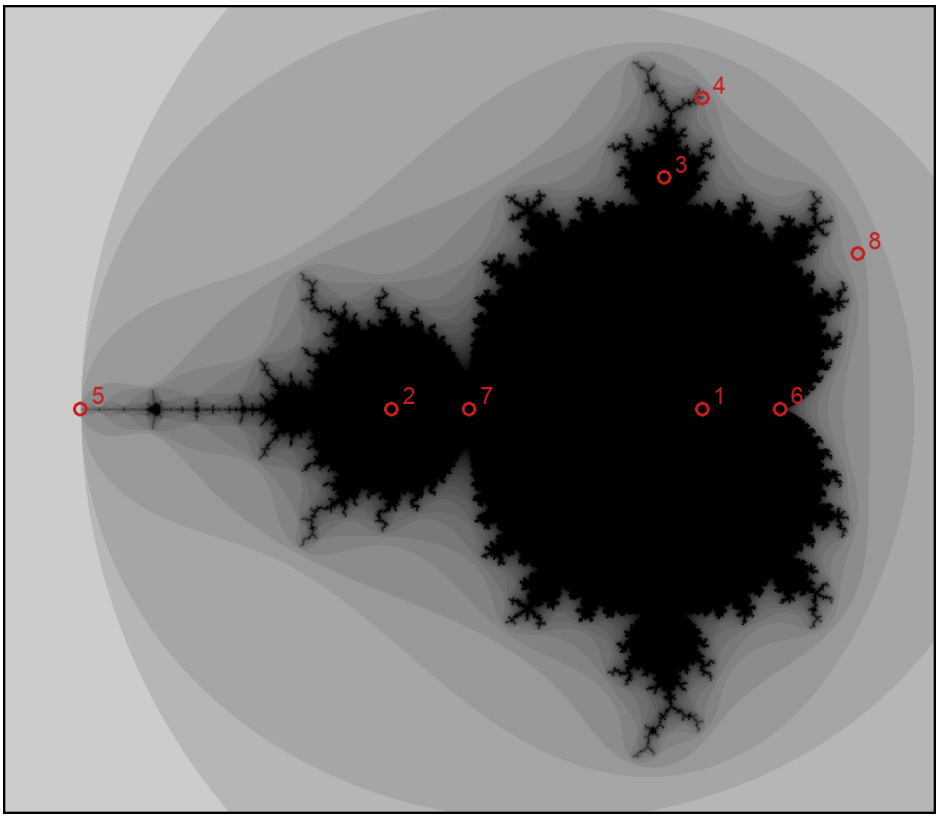


Figure 2: The Mandelbrot set $\mathcal{M}$ in the parameter plane (c-plane) with the eight parameters from Figure 1. Each marked point stands for a whole image in the dynamical plane. Points 1 to 7 lie in $\mathcal{M}$; point 8 lies outside.

What holds instead: $\mathcal{M}$ as a catalogue

The correct relationship is a different one: *each point c* of the parameter plane stands for *a whole* Julia set. Figure 2 shows where the eight parameters of the gallery lie in the parameter plane. One can therefore read $\mathcal{M}$ as a catalogue or table of contents of all quadratic Julia sets [17, 4]:

- If c lies in $\mathcal{M}$, then J_c is connected (points 1 to 7).
- If c lies outside $\mathcal{M}$, then J_c is dust (point 8).

The position within $\mathcal{M}$ also says something about the image. Point 1 is the centre of the large main cardioid, point 2 the centre of the disc $|c + 1| < \frac{1}{4}$ to its left. Point 3 lies at the centre of the "bud" on top of the cardioid; there the critical orbit is periodic with period 3, and, fittingly, in the

rabbit the bubbles meet three at a time. Points 6 and 7 lie exactly on the boundary: at the cusp of the cardioid and at the place where cardioid and disc touch. Points 4 and 5 are tips of fine antennae; there the Julia set no longer has an interior.

Similarity at the boundary: Tan Lei's theorem

There is one place where the idea "Julia set in the Mandelbrot set" has a kernel of truth. Let us consider a parameter c_0 for which the critical orbit, after a preperiod, runs into a cycle without itself passing through 0 again. Examples are $c_0 = i$ with $0 \to i \to -1+i \to -i \to -1+i \to \cdots$ and $c_0 = -2$ with $0 \to -2 \to 2 \to 2 \to \cdots$. Such parameters are called *Misiurewicz points*; they lie on the boundary of $\mathcal{M}$.

In 1990 Tan Lei proved [18]: if one magnifies the Mandelbrot set about a Misiurewicz point c_0 ever more strongly, and likewise the Julia set J_{c_0} about the point $z = c_0$, the two magnified details become ever more similar to each other; in the limit they agree up to a rotation and scaling. So if one zooms into $\mathcal{M}$ at the tip at point 4, one sees the same branchings as in the dendrite J_i at $z = i$. At such boundary points, and specifically in the limit of ever stronger magnification, one does indeed find the Julia set again "in" the Mandelbrot set.

This must be distinguished from the small Apfelmännchen that one finds everywhere in $\mathcal{M}$ when magnifying (Section 3.5). They are scaled-down copies of $\mathcal{M}$ itself, not Julia sets [19].

Try it yourself

1. **By hand.** Choose $c = -1$ and the starting value $z_0 = 0.5$. Use a calculator to compute $z_1, z_2, \ldots, z_6$. As a check: $z_1 = -0.75$ and $z_2 = -0.4375$. Where does the orbit seem to be heading? Compare it with the orbit of 0. Repeat the calculation with $z_0 = 1.5$ and $z_0 = 1.7$. Which of these starting values lies in the basilica K_{-1}, and which does not?

2. **On the computer.** The following program checks whether a starting value z_0 escapes for a fixed c. It uses the escape criterion $|z| > \max(2, |c|)$ and outputs the number of steps until escape, or `None` if the orbit does not escape within `n_max` steps.

```
def escapes(z0, c, n_max=200):
    z = z0
    for n in range(n_max):
        if abs(z) > max(2, abs(c)):
            return n            # escaped after n steps
        z = z*z + c
    return None                 # did not escape

c = -1
for z0 in [0.5, 1.5, 1.7, 0.5+0.5j]:
    print(z0, escapes(z0, c))
print("Does 0 escape?", escapes(0, 0.5+0.5j))
```

   Try other parameters from the gallery. With `escapes(0, c)` you check (within the iteration limit) whether c belongs to the Mandelbrot set; with `escapes(c, c)` you can confirm that the orbits of 0 and c either both escape or both remain bounded.

3. **In the picture.** Place Figure 1 next to Figure 2. For each marker 1 to 8, read off the approximate position of c in the image and check whether it matches the stated value. Which marker lies outside $\mathcal{M}$? For the parameter of point 1, $z = 1$ lies on the Julia set. Does the parameter $c = 1$ still lie within the region shown in Figure 2, and why can it not lie in $\mathcal{M}$?

6 Mandelbrot, Julia and Cantor

What you already know

- A sequence is called bounded if all its terms lie within a fixed circle about the origin (Chapter 2).
- The Mandelbrot set $\mathcal{M}$ consists of the parameters c for which the orbit $0,\ c,\ c^2+c,\ \ldots$ starting at 0 remains bounded (Section 3.3). As soon as a point of the orbit exceeds modulus 2, the orbit escapes.
- For fixed c, the filled Julia set K_c is the set of all starting values z_0 with a bounded orbit, and the Julia set J_c is its boundary. We have $c \in \mathcal{M} \Leftrightarrow K_c$ is connected (Chapter 5).
- In the Koch curve, the middle third of a line segment is replaced; for an exactly self-similar figure made of N copies with reduction factor s, $D = \log N / \log s$ (Sections 4.1 and 4.2).

In this chapter you will find out what becomes of K_c when c does *not* lie in $\mathcal{M}$: the set falls apart into "dust", and this dust is a Cantor set.

In Section 5.4 we established that the filled Julia set K_c is connected if and only if c belongs to the Mandelbrot set. What remained open was what K_c looks like for $c \notin \mathcal{M}$. The answer is surprisingly sharp: there is no intermediate state. Either K_c hangs together in one piece, or the set falls apart into infinitely many separate points, no two of which are connected. To understand this second possibility, we first consider the simplest example of such a "dust set": the Cantor set.

6.1 The middle-thirds Cantor set

6.1.1 Construction

We begin with the interval $C_0 = [0,1]$ and remove the open middle third $\left(\frac{1}{3}, \frac{2}{3}\right)$. Two intervals remain,

$$C_1 = \left[0, \tfrac{1}{3}\right] \cup \left[\tfrac{2}{3}, 1\right].$$

From each of these two intervals we again remove the open middle third and obtain four intervals of length $\frac{1}{9}$:

$$C_2 = \left[0, \tfrac{1}{9}\right] \cup \left[\tfrac{2}{9}, \tfrac{1}{3}\right] \cup \left[\tfrac{2}{3}, \tfrac{7}{9}\right] \cup \left[\tfrac{8}{9}, 1\right].$$

In general, C_n consists of 2^n intervals of length 3^{-n}. The *Cantor set* C is the set of points that are *never* removed in this process, that is, the intersection

$$C = C_0 \cap C_1 \cap C_2 \cap \cdots = \bigcap_{n=0}^{\infty} C_n$$

[20, 21]. Figure 3 shows the first stages.

At first sight, hardly anything seems to remain after infinitely many steps. Certainly all endpoints remain, for instance 0, $\frac{1}{3}$, $\frac{2}{9}$, $\frac{7}{9}$ or 1, because an endpoint, once created, is never removed again. But much more remains, as the representation in base three shows.

6.1.2 The address in base three

Every number $x \in [0,1]$ has a representation in base three (ternary representation)

$$x = 0.a_1a_2a_3\ldots_{(3)} = \frac{a_1}{3} + \frac{a_2}{9} + \frac{a_3}{27} + \cdots, \qquad a_k \in \{0,1,2\}.$$

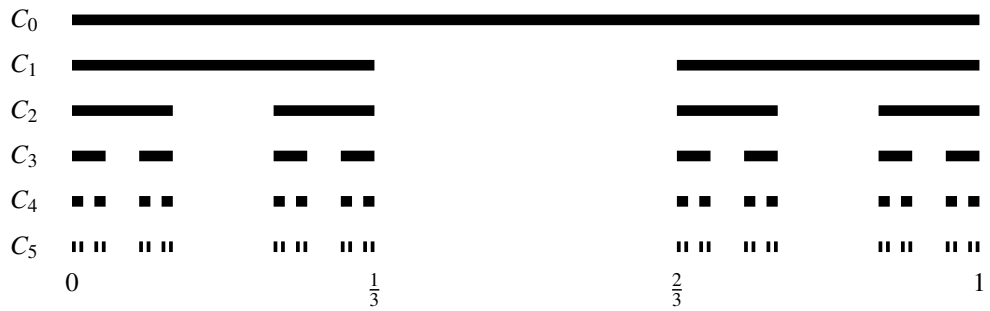


Figure 3: The first six stages $C_0, \ldots, C_5$ of the middle-thirds Cantor set. At each stage the open middle third is removed from every interval; 2^n intervals become 2^{n+1}.

The first digit indicates in which third of $[0, 1]$ the number lies: $a_1 = 0$ for the left, $a_1 = 1$ for the middle, $a_1 = 2$ for the right third. The second digit indicates the third within that third, and so on. Removing the middle thirds therefore means precisely:

> x lies in C if and only if x has a representation in base three that uses only the digits 0 and 2

[21]. The word "has" is important, because some numbers have two representations: $\frac{1}{3} = 0.1_{(3)} = 0.0222\ldots_{(3)}$. The second representation does without the digit 1, so $\frac{1}{3}$ belongs to C; this fits with the fact that $\frac{1}{3}$ is an endpoint. The sequence of digits made of zeros and twos is something like the *address* of a point: at each step it says whether one chooses the left or the right subinterval.

6.1.3 Properties

Uncountably many points. Every infinite sequence of zeros and twos yields a point of C, and different sequences yield different points. The Cantor set therefore has "as many" points as there are infinite 0–2 sequences. These cannot be listed one by one. Suppose we had a complete list $s^{(1)}, s^{(2)}, s^{(3)}, \ldots$ of all such sequences. We then form a new sequence t whose k-th digit is a 2 if the k-th digit of $s^{(k)}$ is a 0, and vice versa. The sequence t differs from every sequence in the list in at least one place and is therefore missing from the list, a contradiction. This is Cantor's *diagonal argument*. The Cantor set is thus *uncountable*, just like the whole interval $[0, 1]$. In particular, it by no means consists only of the countably many endpoints; $\frac{1}{4}$, for example, is not an endpoint and nevertheless lies in C (see the exercises at the end of the chapter).

Total length zero. In the first step an interval of length $\frac{1}{3}$ is removed, in the second two intervals of length $\frac{1}{9}$ are removed, and in the $(n+1)$-th step 2^n intervals of length $3^{-(n+1)}$. The total length removed is a geometric series:

$$\frac{1}{3} + \frac{2}{9} + \frac{4}{27} + \cdots = \sum_{n=0}^{\infty} \frac{2^n}{3^{n+1}} = \frac{1}{3} \cdot \frac{1}{1 - \frac{2}{3}} = 1.$$

So the entire length 1 is removed. Put differently: C_n has total length $\left(\frac{2}{3}\right)^n \to 0$. Although C contains uncountably many points, it has length zero.

No interval, only dust. An interval of length $\ell > 0$ cannot lie in C, because it would have to lie in C_n, whose subintervals are shorter than ℓ for large n. Between any two points of C there

is therefore always a gap. Such a set is called *totally disconnected*: its only connected parts are single points. Nevertheless, no point of C is isolated: in every neighbourhood of a point of C, however small, there lie further points of C (one changes a digit far back in the address). Sets that are closed and have no isolated points are called *perfect*. C is closed because it is the intersection of closed intervals; since C is also bounded, it is called *compact*.

Self-similar. The part of C in $\left[0, \frac{1}{3}\right]$ is a copy of C reduced by the factor 3, and so is the part in $\left[\frac{2}{3}, 1\right]$:

$$C = \tfrac{1}{3}C \;\cup\; \left(\tfrac{1}{3}C + \tfrac{2}{3}\right).$$

In the language of addresses: the left copy consists of the addresses that begin with 0, the right one of those that begin with 2. The Cantor set is thus an exactly self-similar figure made of $N = 2$ copies with reduction factor $s = 3$, like the Koch curve from Section 4.1. The formula from Section 4.2 gives

$$D = \frac{\log 2}{\log 3} \approx 0.6309.$$

The dimension lies between that of a point (0) and that of a line segment (1). In Chapter 7 we show that the Hausdorff dimension of the Cantor set also has this value.

> **Historical context** Georg Cantor published the middle-thirds set in 1883 in the fifth part of his work on infinite linear point sets [20], as an example of a perfect set that contains no interval. In 1910 L. E. J. Brouwer proved that every non-empty, compact, perfect and totally disconnected set in the plane is topologically equivalent to the middle-thirds set [22]: they can be mapped onto each other continuously with a continuous inverse. This is why all such sets are called *Cantor sets*, even if they look quite different.

6.2 The Fatou–Julia dichotomy

For $f_c(z) = z^2 + c$, 0 is the only *critical point*, that is, the only point with $f_c'(z) = 2z = 0$. Its orbit $0,\ c,\ c^2 + c,\ \ldots$ is called the *critical orbit*. By Section 3.3, c belongs to $\mathcal{M}$ if and only if this orbit remains bounded. The following result goes back in essence to Pierre Fatou and Gaston Julia [10, 12, 11].

Theorem (dichotomy).

(a) If the critical orbit is bounded, that is, $c \in \mathcal{M}$, then K_c and J_c are connected.

(b) If the critical orbit escapes, that is, $c \notin \mathcal{M}$, then $K_c = J_c$, and this set is a Cantor set: compact, perfect and totally disconnected.

[10, 23, 24]

Dichotomy means division into two: there are exactly two possibilities and nothing in between. In particular, for $z^2 + c$ there is no filled Julia set that consists of exactly two or exactly a hundred pieces. The Mandelbrot set thus sorts the Julia sets into two classes; this explains the name *connectedness locus* from Section 5.4. In case (b) one speaks descriptively of *Cantor dust* or *Fatou dust*.

6.2.1 Why this is so: the idea of the proof

The idea of the proof can be described without calculation. It rests on an observation about computing backwards. To get from an image point w to the preimages z with $z^2 + c = w$, one has to compute

$$z = \pm\sqrt{w - c}.$$

Every point $w \neq c$ therefore has two preimages, while the point $w = c$ has only one, namely $z = 0$. This is the reason why the critical point plays a special role.

We now take a large, roundish region V (a disc, say) that contains K_c, and compute its preimage $f_c^{-1}(V)$, that is, the set of all z that land in V. There are two cases.

- *The point c lies in V.* Then $V - c$ encloses the origin, and the two branches of the root $+\sqrt{\ }$ and $-\sqrt{\ }$ merge into each other. The preimage is again *one* connected region.
- *The point c does not lie in V.* Then the two branches of the root separate, and the preimage falls apart into *two* separate regions U_0 and U_1, each of which is mapped by f_c onto the whole of V.

Repeating the backward computation gives a sequence of nested regions whose intersection is K_c.

If the critical orbit is bounded, all its points lie in K_c and hence in each of the regions. The first case occurs at every step; one obtains a chain of nested connected regions, and their intersection K_c is connected.

If the critical orbit escapes, V can be chosen so that V contains the set K_c but not the point c, and so that U_0 and U_1 again lie in V. Then the preimage falls apart into two parts, their preimages into four parts, then eight, sixteen, ... This is exactly the pattern of the Cantor construction. One also shows that the parts become smaller at every step and shrink down to single points [10, 23]. Every point of J_c thereby receives an address of zeros and ones that states whether its orbit lies in U_0 or in U_1 at each step, and every such address belongs to exactly one point. The map f_c acts on these addresses by deleting the first digit [24, 25]. The same happens for the middle-thirds set: the maps $x \mapsto 3x$ on $\left[0, \frac{1}{3}\right]$ and $x \mapsto 3x - 2$ on $\left[\frac{2}{3}, 1\right]$ delete the first digit of the ternary address. For $c \notin \mathcal{M}$ the dynamics of f_c on J_c is therefore essentially the same as on the Cantor set.

6.3 The real case $c < -2$

For real parameters $c < -2$ the disintegration can be followed on the number line. Since $\mathcal{M} \cap \mathbb{R} = \left[-2, \frac{1}{4}\right]$, these parameters lie outside $\mathcal{M}$ [10]. We calculate with the example $c = -2.5$, that is, $f(x) = x^2 - 2.5$.

The large interval. The larger fixed point of f is the solution of $x^2 + c = x$, namely

$$\beta = \frac{1 + \sqrt{1 - 4c}}{2} = \frac{1 + \sqrt{11}}{2} \approx 2.158.$$

For $|x| > \beta$ we have $f(x) > \beta$, and the orbit grows beyond all bounds. Everything of interest therefore takes place in the interval $I = [-\beta, \beta]$.

Step 1: two intervals. The graph of f is an upward-opening parabola with vertex $(0, c)$. Since $c < -\beta$, the middle of I is mapped below $-\beta$ and escapes in the next step. More precisely,

$f(x) < -\beta$ if and only if $x^2 < -\beta - c \approx 0.342$, that is, for $|x| < 0.585$. Two separate intervals remain,

$$I_0 \approx [-2.158, -0.585], \qquad I_1 \approx [0.585, 2.158].$$

Each of them is mapped by f strictly monotonically onto the whole interval I: I_0 decreasingly, I_1 increasingly.

Step 2: four intervals. Because f maps each of the intervals I_0 and I_1 onto the whole of I, each of them again contains an inner piece whose points fall into the gap $(-0.585, 0.585)$ in the next step. What remains are four intervals,

$$[-2.158, -1.756], \quad [-1.384, -0.585], \quad [0.585, 1.384], \quad [1.756, 2.158].$$

Step n: 2^n intervals. And so it continues: after n steps there remain 2^n intervals whose points stay in I for at least n iterations (Figure 4). The points that stay in I *for ever* form the intersection of all stages, a Cantor set $\Lambda \subset \mathbb{R}$. All other real starting values escape. One can show that all non-real starting values escape as well; hence $K_c = J_c = \Lambda$ [25, 10]. For c close to -2, proving that the intervals really do shrink down to points takes somewhat more work than for strongly negative c.

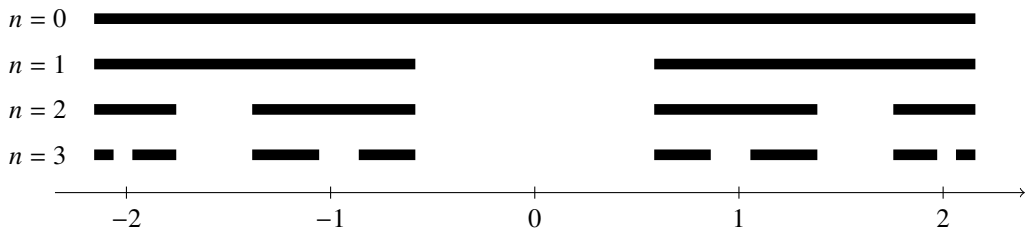


Figure 4: The real case $c = -2.5$: starting values whose orbit stays in the interval $I = [-\beta, \beta]$, $\beta \approx 2.158$, for at least 0, 1, 2 or 3 steps respectively. Unlike in the middle-thirds set, the gaps differ in size, because the parabola does not stretch equally strongly everywhere; the pattern "every interval falls apart into two" is, however, the same.

The limiting case $c = -2$. For $c = -2$ we have $\beta = 2$, and the vertex lies exactly at $-2 = -\beta$. Nothing drops out any more: the two intervals $[-2, 0]$ and $[0, 2]$ touch at 0, and $J_{-2} = K_{-2} = [-2, 2]$, the line segment from Section 5.2. An arbitrarily small step to the left of $c = -2$ makes the segment fall apart into dust.

The set Λ is a Cantor set in Brouwer's sense, but not the middle-thirds set: its pieces are not reduced exactly to scale but slightly distorted. Its dimension is in general not $\log 2/\log 3$ but depends on c (Chapter 7).

6.4 The transition in the plane

Figure 5 shows what happens when the parameter at the tip of the main body of $\mathcal{M}$, at $c = \frac{1}{4}$, is pushed to the right out of $\mathcal{M}$. For $c = 0.25$, K_c is the connected "cauliflower" from Section 5.2. Already for $c = 0.26$ the critical orbit escapes; however, it exceeds modulus 2 only after 30 steps. By the theorem, $K_{0.26}$ is therefore Cantor dust, even though at first sight the picture looks almost the same as before. The dark areas, including the black patches, do not belong to K_c; they consist of starting values that escape only very slowly and therefore have not yet escaped

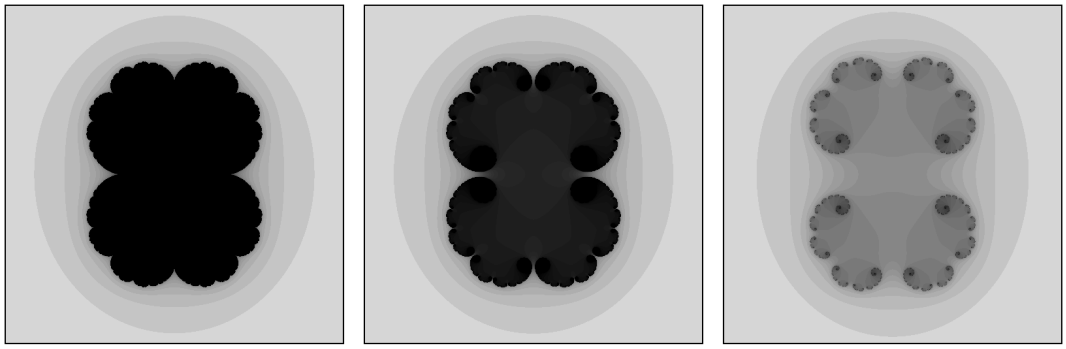


Figure 5: Filled Julia sets for $c = 0.25$ (in $\mathcal{M}$, connected), $c = 0.26$ (just outside $\mathcal{M}$) and $c = 0.35$ (further outside). Grey levels indicate the escape time: light means fast, dark slow escape, black no escape within the iteration limit. At $c = 0.26$, at finite resolution, the set still looks almost connected; in fact it is totally disconnected.

within the iteration limit. At $c = 0.35$ the critical orbit escapes after only 8 steps, and all that remains of the cauliflower are scattered spirals of dust grains.

The picture shows a limitation of all computer graphics: a pixel has a finite size, and a finite number of iterations cannot distinguish a very slow escape from remaining bounded. Whether a Julia set is connected is therefore decided not from the picture but with the theorem: one checks the *one* critical orbit.

6.5 Mandelbrot, Julia and Cantor: the connection

This brings us full circle.

- Every point c of the parameter plane stands for a Julia set J_c in the dynamical plane. The Mandelbrot set is a kind of table of contents of these Julia sets.
- If c lies in $\mathcal{M}$, then J_c is connected: a circle, a basilica, a rabbit, a dendrite, a line segment.
- If c lies outside $\mathcal{M}$, then J_c is a Cantor set. All these Julia sets are topologically equivalent to the middle-thirds set, but look very different and can have different dimensions.
- The decision is made by a single orbit, that of the critical point 0. It is precisely this orbit that the Apfelmännchen program from Section 3.3 computes.

Without saying so, the 1990 program therefore answered, approximately and for every pixel, the question: "Does the corresponding Julia set fall apart into dust or not?" How to measure the "size" of such dust sets, which cannot be captured meaningfully by either a length or an area, is the subject of Chapter 7.

Try it yourself

1. *What lies in the Cantor set?* Calculate the representations of $\frac{1}{4}$, $\frac{1}{3}$ and $\frac{1}{2}$ in base three. (Hint: multiply by 3 again and again and write down the integer part as the next digit.) You should obtain $\frac{1}{4} = 0.020202\ldots_{(3)}$, $\frac{1}{3} = 0.1_{(3)} = 0.0222\ldots_{(3)}$ and $\frac{1}{2} = 0.1111\ldots_{(3)}$. Which of the three numbers lie in C? Which of them is not an endpoint of a removed interval? Check your

result for $\frac{1}{2}$ directly against Figure 3 as well.

2. *How much is removed?* Calculate the total length of $C_1, \ldots, C_5$ and the length removed after five steps. Confirm with the formula for the geometric series that in total the length 1 is removed. How many steps does it take until less than 1 % of the length remains?

3. *Counting dust.* For $c = -2.5$, the following program lays a fine grid of starting values on $[-2, 2]$ and counts how many still lie in $I = [-\beta, \beta]$ after 1,2,3, ... iterations.

```
c = -2.5
beta = (1 + (1 - 4*c) ** 0.5) / 2      # right fixed point, approx. 2.158
N = 100000                              # grid points on [-2, 2]
points = [-2 + 4*j/N for j in range(N + 1)]
for n in range(1, 11):
    points = [x*x + c for x in points]
    points = [x for x in points if abs(x) <= beta]
    print(n, len(points), round(len(points) / (N + 1), 4))
```

After one step about 71 % remain, after ten steps only around 6 %. Calculate the ratio of successive counts. What does it mean that it is roughly constant? Replace c by -2.1 and by -2 and compare.

7 The Hausdorff dimension

What you already know

In Section 4.2 you met the *similarity dimension*: if a figure consists of N copies of itself, each scaled down by the factor s, then $N = s^D$, so $D = \log N/\log s$. For the Koch curve (Section 4.1) this gave $D = \log 4/\log 3 \approx 1.26$. You also know the box-counting method from there: count squares of side length ε and examine $\log N(\varepsilon)/\log(1/\varepsilon)$. In Chapter 6 you built the Cantor set: from $[0,1]$ the open middle third of every piece is removed again and again. And in Chapters 5 and 6 you saw that Julia sets J_c can be connected or Cantor dust, depending on whether c lies in the Mandelbrot set $\mathcal{M}$.

The similarity dimension from Section 4.2 has a catch: it is defined only for figures that consist exactly of scaled-down copies of themselves. The boundary of the Mandelbrot set and the path of a randomly wandering particle are not such figures. This chapter introduces the notion of dimension that mathematics uses for *all* sets: the Hausdorff dimension. It generalises the similarity dimension, agrees with it in the nice cases and allows statements that could not even be formulated without it, for instance: "The boundary of $\mathcal{M}$ is two-dimensional."

7.1 The problem: length zero is not the same as length zero

How large is the Cantor set C? After the first step two intervals of length $\frac{1}{3}$ remain, after the second four of length $\frac{1}{9}$, after k steps 2^k intervals of length 3^{-k}. The total length is therefore

$$2^k \cdot 3^{-k} = \left(\tfrac{2}{3}\right)^k \longrightarrow 0 \qquad (k \to \infty).$$

The Cantor set has length 0, just like a single point. Nevertheless it is anything but small: it contains uncountably many points (Chapter 6), and every piece of it, however small, again looks like the whole.

The familiar measures are too coarse here. For such sets, length knows only two answers, "zero" or "positive"; and so does area. The Koch curve slips through the grid the other way: its length is infinite, its area zero. What is wanted, then, is a yardstick that resolves more finely than "point, line, surface" and that can distinguish between a point and the Cantor set.

7.2 The idea: covering with small sets

One measures a set F by covering it with small pieces $U_1, U_2, \ldots$, for example with intervals, discs or squares. Of each piece, only its *diameter* $\operatorname{diam} U_i$ counts, that is, the greatest distance between two of its points. The pieces may be of different sizes, there may be (countably) infinitely many of them, but none may be larger than a prescribed bound δ.

For a line segment one would add up the lengths of the pieces, for a surface roughly the squares of the diameters. In general one chooses an exponent $t \geq 0$ and forms the sum

$$\sum_i (\operatorname{diam} U_i)^t.$$

(We write t because s was already the scaling factor in Section 4.2; in the specialist literature the exponent is usually called s.) Now one looks for the most economical covering and then lets δ tend to 0:

$$\mathcal{H}^t_\delta(F) = \inf\Big\{\sum_i (\operatorname{diam} U_i)^t \,:\, F \subseteq \bigcup_i U_i,\ \operatorname{diam} U_i \leq \delta\Big\}, \qquad \mathcal{H}^t(F) = \lim_{\delta \to 0} \mathcal{H}^t_\delta(F).$$

The "inf" (infimum) means: the largest number that no admissible covering undercuts, intuitively the sum for the best covering. The limit exists because $\mathcal{H}^t_\delta(F)$ can only grow as δ is made smaller: fewer coverings are then allowed. The condition $\delta \to 0$ is essential. If large pieces were permitted, one would cover the whole Cantor set with a single interval of length 1 and not see its fine structure at all.

The number $\mathcal{H}^t(F)$ is called the *t-dimensional Hausdorff measure* of F. For integer t it is an old acquaintance: $\mathcal{H}^0$ counts points, $\mathcal{H}^1$ is length, and $\mathcal{H}^2$ is area up to a fixed factor [21].

Two trial runs. A single point fits into an arbitrarily small set of diameter η. For $t > 0$, η^t is arbitrarily small, so $\mathcal{H}^t(\text{point}) = 0$; only for $t = 0$ does one obtain 1.

A line segment of length 1 is divided into n sub-segments of length $\frac{1}{n}$. The sum is

$$n \cdot \left(\frac{1}{n}\right)^t = n^{1-t} \longrightarrow \begin{cases} \infty, & t < 1, \\ 1, & t = 1, \\ 0, & t > 1, \end{cases} \qquad (n \to \infty).$$

For $t > 1$ the segment is "too thin", for $t < 1$ "too large". Only with the exponent $t = 1$ does a sensible number come out: the length. (That for $t < 1$ there is no cleverer covering that keeps the sum small still has to be shown; but it is true.)

7.3 The critical value

What happens for the line segment holds for every set: there is exactly one value t_0 at which $\mathcal{H}^t(F)$ jumps from ∞ to 0. For $t < t_0$, $\mathcal{H}^t(F) = \infty$; for $t > t_0$, $\mathcal{H}^t(F) = 0$. The reason is a short calculation: if $\operatorname{diam} U_i \le \delta$ and $t' > t$, then $(\operatorname{diam} U_i)^{t'} \le \delta^{t'-t} (\operatorname{diam} U_i)^t$. So if $\mathcal{H}^t(F)$ is finite, $\mathcal{H}^{t'}(F)$ is pushed down by the factor $\delta^{t'-t} \to 0$ and is zero.

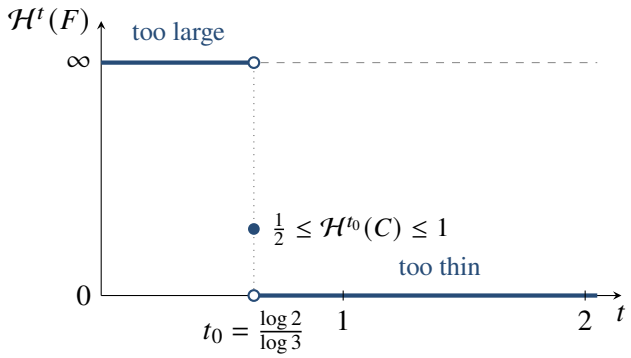


Figure 6: The t-dimensional Hausdorff measure of the Cantor set C as a function of t: to the left of the critical value it is infinite, to the right of it zero. Only exactly at the critical value can a positive, finite number come out. This jump value is the Hausdorff dimension.

Definition. The jump value is called the *Hausdorff dimension* of F:

$$\dim_{\mathrm{H}} F = \inf\{t : \mathcal{H}^t(F) = 0\} = \sup\{t : \mathcal{H}^t(F) = \infty\}.$$

At the jump value itself, $\mathcal{H}^{t_0}(F)$ can be zero, positive and finite, or infinite; the dimension says nothing about this [21].

The Cantor set. After k construction steps, C is covered by 2^k intervals of length 3^{-k} (Figure 7). The sum is

$$2^k \cdot \left(3^{-k}\right)^t = \left(\frac{2}{3^t}\right)^k.$$

If $3^t > 2$, this tends to 0; if $3^t < 2$, it grows without bound. The changeover lies at $3^t = 2$, that is, at

$$t_0 = \frac{\log 2}{\log 3} \approx 0.6309.$$

Strictly speaking, this calculation shows only $\mathcal{H}^t(C) = 0$ for $t > t_0$, because it uses one particular covering. For $t < t_0$ one still has to justify that no other covering is substantially more economical. That is the harder part. The idea: an arbitrary small interval meets at most a few stage-k intervals of suitable size, and it can be replaced by these without much loss. The result is $\frac{1}{2} \le \mathcal{H}^{t_0}(C) \le 1$ [21, Ch. 3], hence $\dim_{\mathrm{H}} C = \log 2/\log 3$.

This solves the problem from the beginning of the chapter: a point has dimension 0, the Cantor set dimension 0.63. Both have length 0, but the Cantor set is "larger" in a precisely measurable sense.



Figure 7: The first stages of the Cantor set. Each stage is at the same time a covering of C: 2^k intervals of length 3^{-k}.

Countable sets have dimension 0. If the points of a set lie in a list $x_1, x_2, x_3, \ldots$, one covers x_k with an interval of length $\delta \cdot 2^{-k}$. For $t > 0$ the sum is at most $\delta^t \left(2^{-t} + 2^{-2t} + 2^{-3t} + \ldots\right)$, a geometric series times δ^t, and hence arbitrarily small. That is why, for example, the rational numbers have Hausdorff dimension 0, even though they lie densely everywhere on the number line. The Cantor set, by contrast, is uncountable, and its dimension is positive.

> **Historical context** Felix Hausdorff introduced these measures and fractional dimension in 1918 in the paper "Dimension und äußeres Maß" [26]. The volume of the *Mathematische Annalen* bears the date 1918/1919, which is why both years are found in the literature. His fully worked example was already the Cantor set with dimension $\log 2/\log 3$. Abram S. Besicovitch developed the theory systematically from the late 1920s onwards [27]; this is why one also speaks of the Hausdorff–Besicovitch dimension. Mandelbrot's word "fractal" was added only decades later [1].

7.4 Hausdorff dimension and similarity dimension

The calculation for the Cantor set was essentially the formula $N = s^D$ from Section 4.2 with $N = 2$ and $s = 3$. This is no coincidence. Let a self-similar set F consist of m copies that are scaled down by the factors $r_1, \ldots, r_m < 1$. Its similarity dimension is the solution t of the *Moran equation*

$$r_1^t + r_2^t + \cdots + r_m^t = 1.$$

If all factors are equal to $r = 1/s$, this reads $m \cdot s^{-t} = 1$, so $m = s^t$: exactly the formula from Section 4.2. The Moran equation, however, also allows unequal parts. If, for instance, one keeps the pieces $[0, \frac{1}{2}]$ and $[\frac{3}{4}, 1]$ of $[0, 1]$ and repeats this, it reads $(\frac{1}{2})^t + (\frac{1}{4})^t = 1$. With $x = (\frac{1}{2})^t$ this becomes $x + x^2 = 1$, so $x = \frac{\sqrt{5}-1}{2}$ and $t = \log \frac{1+\sqrt{5}}{2} \,/\, \log 2 \approx 0.694$.

Theorem (Moran 1946, Hutchinson 1981). If the copies do not overlap substantially, then the Hausdorff dimension and the box dimension are equal to the similarity dimension [28, 29], [21, Sect. 9.2].

"Not overlapping substantially" is made precise by the *open set condition*: there is a non-empty open set V whose scaled-down copies $S_1(V), \ldots, S_m(V)$ again lie in V and are pairwise disjoint. Contact at the boundary is allowed. The four parts of the Koch curve meet at single points, the three parts of the Sierpiński triangle at corners; both are harmless. For all the examples from Section 4.2, one may therefore use the convenient formula $D = \log N / \log s$ and obtains the Hausdorff dimension.

When the copies overlap. The segment $[0, 1]$ is the union of the three half-length segments $[0, \frac{1}{2}]$, $[\frac{1}{4}, \frac{3}{4}]$ and $[\frac{1}{2}, 1]$. Applied blindly, the formula yields $\log 3 / \log 2 \approx 1.58$, more than the dimension 1 of the segment. The mistake: the middle copy lies almost entirely on the other two, so points have been counted twice. With overlaps, the dimension can therefore be smaller than the similarity dimension. When exactly this happens is a current research question (Chapter 9).

7.5 Box counting: the computation-friendly sister

In the box counting of Section 4.2, all pieces must be of the *same* size: one counts the smallest number $N_\delta(F)$ of sets of diameter at most δ that are needed, and examines $\log N_\delta(F) / \log(1/\delta)$ as $\delta \to 0$. For the Hausdorff dimension, by contrast, the pieces may be of different sizes, and they may be fitted to the set individually. Because this gives more freedom, it is always the case that

$$\dim_{\mathrm{H}} F \;\le\; \dim_{\mathrm{Box}} F$$

(provided the limit defining the box dimension exists; otherwise one takes the lower limit) [21]. For self-similar sets satisfying the open set condition the two are equal. In general they are not:

Example. The set $F = \{0\} \cup \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\}$ is countable, so $\dim_{\mathrm{H}} F = 0$. For box counting with boxes of width ε, however, it looks different. The distance between $\frac{1}{n}$ and $\frac{1}{n+1}$ is $\frac{1}{n(n+1)} \approx \frac{1}{n^2}$. As long as it is greater than ε, that is, roughly for $n < 1/\sqrt{\varepsilon}$, each of these points needs a box of its own: about $1/\sqrt{\varepsilon}$ of them. All remaining points lie so densely in the interval $[0, \sqrt{\varepsilon}\,]$ that this has to be covered entirely: another roughly $\sqrt{\varepsilon}/\varepsilon = 1/\sqrt{\varepsilon}$ boxes. Hence $N(\varepsilon) \approx 2\,\varepsilon^{-1/2}$ and

$$\dim_{\mathrm{Box}} F = \frac{1}{2} \neq 0 = \dim_{\mathrm{H}} F \qquad [21, \text{Ch. 2}].$$

More drastic still: the rational numbers in $[0, 1]$ lie densely, so every covering by boxes has to cover the whole interval. Their box dimension is 1, their Hausdorff dimension 0.

Why use box counting at all, then? Because it is easy to compute: a computer lays a grid over an image, counts the occupied boxes and fits a straight line in a log–log plot. The Hausdorff dimension, for its part, is the mathematically better quantity. In particular, the dimension of a countable union of sets is the supremum of the individual dimensions, and box counting does not satisfy this, as the example of the rational numbers shows.

7.6 A table of values

Table 1 sets well-known sets side by side. The columns for length and area show the jump rule from Section 7.3: if the dimension is less than 1, the length is zero; if it is greater than 1, the length is infinite; likewise for area with the threshold 2.

Set	Structure	$\dim_{\mathrm{H}}$	Length $\mathcal{H}^1$	Area
Point	–	0	0	0
Line segment of length 1	3 copies, factor $\frac{1}{3}$	1	1	0
Cantor set (Ch. 6)	2 copies, factor $\frac{1}{3}$	$\frac{\log 2}{\log 3} \approx 0.6309$	0	0
Koch curve (Sect. 4.1)	4 copies, factor $\frac{1}{3}$	$\frac{\log 4}{\log 3} \approx 1.2619$	∞	0
Sierpiński triangle	3 copies, factor $\frac{1}{2}$	$\frac{\log 3}{\log 2} \approx 1.5850$	∞	0
Square	9 copies, factor $\frac{1}{3}$	2	∞	positive
Path of a Brownian motion in the plane	random	2 (almost surely) [30]	∞	0
Boundary of the Mandelbrot set $\partial\mathcal{M}$	no exact self-similarity	2 [31]	∞	unknown

Table 1: Hausdorff dimension, length and area of some sets. The first six values follow from the Moran equation with the open set condition [21]. The Brownian path is the trace of a randomly wandering particle; that its area is zero goes back to Paul Lévy [21, Ch. 16].

Two rows stand out. The Brownian path is a *curve*, it can be traversed in one go, and yet it has dimension 2 like a surface, without having any area. And for the boundary of the Mandelbrot set the area is still unknown today; this is the subject of the next section.

7.7 Dimension in complex dynamics

Julia sets for small c. For $c = 0$, J_0 is the unit circle (Chapter 5); its dimension is 1. If c lies in the main cardioid of $\mathcal{M}$, J_c is still a closed curve without self-contact, but a crumpled one. Ruelle proved in 1982 that the dimension $\dim_{\mathrm{H}} J_c$ depends very smoothly (even real-analytically) on c there [32]. In the specialist literature the approximation

$$\dim_{\mathrm{H}} J_c \approx 1 + \frac{|c|^2}{4 \ln 2}$$

is given for small $|c|$ and is usually attributed to Ruelle; whether it already appears in this form in his original paper is unclear. For $c = 0.1$ it would give about 1.0036. So for small $c \neq 0$ the Julia set is "a tiny bit more than a curve": even a minute perturbation of z^2 turns the smooth circle into a fractal.

Cantor dust. If c lies outside $\mathcal{M}$, then J_c is a Cantor dust (Chapter 6), and its dimension tends to 0 as $|c|$ becomes ever larger: the dust becomes ever thinner.

The boundary of the Mandelbrot set. The Mandelbrot set itself has interior points, for example the whole main cardioid. Its dimension is therefore trivially 2. Its *boundary* $\partial\mathcal{M}$, on the other hand, does not contain a single disc, however small. One would expect it to behave like

a curve, perhaps with a dimension slightly above 1, as the boundary of the Koch snowflake does with 1.26.

Theorem (Shishikura). $\dim_{\mathrm{H}} \partial\mathcal{M} = 2$. Moreover, $\dim_{\mathrm{H}} J_c = 2$ for "typical" parameters c on $\partial\mathcal{M}$ (more precisely: for a set of boundary points that is large in the topological sense, a so-called residual set) [31].

What does this mean? The boundary of $\mathcal{M}$ is so heavily branched and crumpled that, in the sense of dimension, it is as large as a piece of the plane, even though it does not "fill" any area. By the jump rule, the measure $\mathcal{H}^t(\partial\mathcal{M})$ is infinite for every $t < 2$; in particular the boundary has infinite length, and indeed "in every dimension below 2". The magnified details from Section 3.5, in which new branchings, spirals and small copies of $\mathcal{M}$ appear at every point, give an impression of this. More dimension than this is not possible in the plane.

The idea of the proof can be sketched in words. Shishikura studies parameters at which a periodic point is just becoming "parabolic", as at $c = \frac{1}{4}$ at the cusp of the main cardioid. If c is shifted slightly there, subsets arise in the Julia sets whose dimension comes arbitrarily close to 2. He then transfers this fine structure to the parameter plane, that is, to the boundary of $\mathcal{M}$. That the dynamical plane and the parameter plane mirror each other so closely fits with what was said in Chapter 5 about $\mathcal{M}$ as a "catalogue" of the Julia sets.

An important qualification: dimension 2 does *not* mean positive area. The Brownian path, too, has dimension 2 and area 0. Whether $\partial\mathcal{M}$ has positive area is open (Chapter 9); Shishikura's theorem shows that dimension alone cannot decide this question. Dimension 2 also occurs for Julia sets of other functions, for example for λe^z [33] (Chapter 8).

> **Historical context** Shishikura's result circulated as a preprint from 1991, a short version appeared in 1994, and the complete paper in 1998 in the *Annals of Mathematics* [31], that is, only after the original 1990 version of this paper. It settles in retrospect the question that was left open there (Section 4.2): not only $\mathcal{M}$ but also its boundary has dimension 2.

7.8 Why this matters

Size beyond length and area. The Cantor set, the rational numbers, the Koch curve and the Brownian path all have area 0. Their dimensions 0.63, 0, 1.26 and 2 nevertheless clearly distinguish them [21]. The Hausdorff dimension is the tool with which one brings order to the "null sets".

Measuring exceptional sets. Many theorems hold "almost everywhere", that is, except on a set of measure zero. The dimension says how large these exceptions are. An example from number theory: the numbers in $[0, 1]$ in whose decimal expansion the digit 7 never occurs form a set of length 0. But it is built self-similarly from 9 copies with the factor $\frac{1}{10}$, and so has dimension $\log 9/\log 10 \approx 0.954$, almost 1. The Cantor set is the same thing for base three and the digit 1 [21, Ch. 10].

Scaling laws in nature and technology, with caution. Lewis F. Richardson measured coastlines with rulers of length δ and found that the measured length grows like δ^{1-D} as δ becomes smaller; for the west coast of Britain he found $D \approx 1.25$ [34]. In 1967 Mandelbrot interpreted D as a fractional dimension [35]. Similar power laws occur in growth processes, turbulence or porous materials, and in medicine branched structures such as blood vessels or

bronchi are described in this way. Here one has to remain honest: a real coastline is not a mathematical set, and measured data cover only a few orders of magnitude. A D obtained in this way is an empirical box-counting exponent with model assumptions, not a proven value of a Hausdorff dimension [21].

A solved puzzle. How convoluted is the outer boundary of the trace of a random walker in the plane? On the basis of computer images, Mandelbrot conjectured the dimension $\frac{4}{3}$. Lawler, Schramm and Werner proved this in 2001 [36]; more on this in Chapter 9.

Try it yourself

1. **Box counting by hand.** Draw the Koch curve up to the third stage on squared paper, with a base segment of 27 squares. Lay grids with box width 9, 3 and 1 over it (that is, $\varepsilon = \frac{1}{3}, \frac{1}{9}, \frac{1}{27}$ of the base segment) and in each case count the boxes that the curve touches. By what factor does the number grow when ε shrinks to a third? Compare $\log(\text{factor})/\log 3$ with $\log 4/\log 3 \approx 1.26$. Why is the agreement not perfect?
2. **A fractal of your own.** Divide a square into 3×3 equal sub-squares, keep only the four corner squares and repeat this in each remaining square ("Cantor dust in the square"). Use $N = s^D$ to calculate the dimension. You obtain $\log 4/\log 3$, the same value as for the Koch curve, even though one set is a connected curve and the other is dust. So what does the dimension *not* tell you about a set? Variant: also keep the centre square (5 of 9), or in $[0, 1]$ keep the pieces $[0, \frac{1}{2}]$ and $[\frac{3}{4}, 1]$ and solve the Moran equation from Section 7.4.
3. **Box dimension of the Cantor set on the computer.** The program generates the 2^{15} left endpoints of the stage-15 intervals and counts how many boxes of width 2^{-m} are occupied. It prints m, N, the quotient $\log N/\log(1/\varepsilon)$ and the slope in the log–log plot between ε and $\varepsilon/16$.

```
from math import log
pts = [0.0]                       # left ends of the Cantor intervals
for k in range(15):
    pts = [x / 3 for x in pts] + [x / 3 + 2 / 3 for x in pts]
N = {}
for m in range(2, 19):
    eps = 2 ** -m                 # box width 1/2^m
    N[m] = len({int(x / eps) for x in pts})    # occupied boxes
    q = log(N[m]) / log(1 / eps)
    if m >= 6:
        s = log(N[m] / N[m - 4]) / log(16)     # slope in the log-log plot
        print(m, N[m], round(q, 3), round(s, 3))
```

For $m = 18$ the quotient gives only 0.683, whereas the slope gives 0.625, close to the exact value 0.6309. Think about why the quotient falls so slowly (hint: $N \approx K \cdot \varepsilon^{-D}$ with a constant factor K) and why the slope fluctuates (hint: the base-two grid does not match the base-three structure of the set). What happens with box widths 3^{-m}?

8 Beyond $z^2 + c$

What you already know

- The polar form $z = r(\cos\varphi + i\sin\varphi)$: when multiplying, the moduli are multiplied and the angles added (Section 1.6.2).
- $\mathcal{M}$ is the set of c for which the orbit of 0 under $f_c(z) = z^2 + c$ remains bounded; an escape radius of 2 suffices (Section 3.3).
- K_c is connected if and only if the orbit of 0 remains bounded; otherwise J_c is a Cantor dust (Chapters 5 and 6).
- The Hausdorff dimension also measures sets that have neither length nor area (Chapter 7).

The formula $z^2 + c$ is only the beginning. The 1990 Facharbeit already hinted at the end of Chapter 3 that other functions can be iterated as well. This chapter follows that path: to higher powers, to general polynomials and rational functions, and finally to the transcendental functions e^z, $\sin z$, $\cos z$, $\tan z$ and $\ln z$. Along the way it becomes clear which parts of the theory from Chapters 5 and 6 survive and which do not.

8.1 Higher powers: $z^d + c$ and Multibrot sets

For a natural number $d \geq 2$ we consider

$$f_{d,c}(z) = z^d + c.$$

The derivative $d\,z^{d-1}$ vanishes only at $z = 0$. So, as for $z^2 + c$, there is exactly one critical point, and everything depends on its orbit 0, c, $c^d + c$, … The dichotomy from Chapter 6 holds unchanged: if the orbit of 0 remains bounded, the filled Julia set is connected; otherwise it is a Cantor dust [10]. The set

$$\mathcal{M}_d = \{\, c \in \mathbb{C} \mid \text{the orbit of 0 under } f_{d,c} \text{ is bounded} \,\}$$

is called the *Multibrot set* of degree d; $\mathcal{M}_2 = \mathcal{M}$. The escape radius 2 also remains usable: the estimate from Section 3.3 works word for word, because $|z|^d \geq |z|^2$ holds for $|z| > 2$.

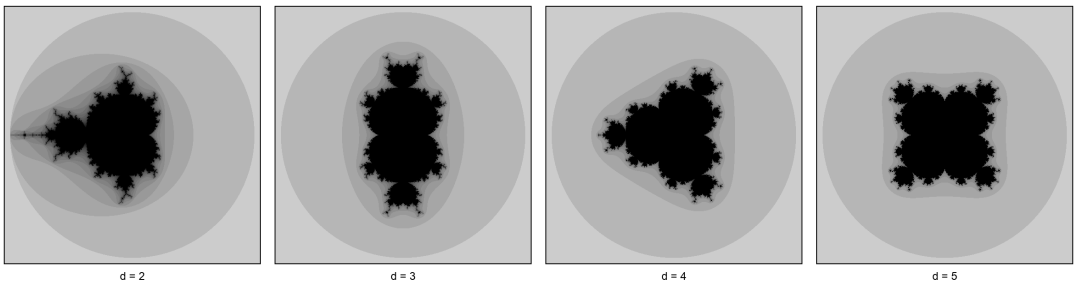


Figure 8: The Multibrot sets $\mathcal{M}_d$ for d = 2,3,4,5 in the parameter plane. Black: the orbit of 0 has not escaped within the iteration limit. Shades of grey: escape time as in Section 3.4. $\mathcal{M}_3$ is point-symmetric (twofold), $\mathcal{M}_4$ has threefold and $\mathcal{M}_5$ fourfold symmetry.

Why $d-1$ and not d?

Figure 8 shows a surprising count: $\mathcal{M}_d$ has $(d-1)$-fold rotational symmetry, whereas the associated Julia sets have d-fold symmetry. Both follow from a short rotation argument using the polar form. Let $\omega = \cos\frac{2\pi}{m} + i\sin\frac{2\pi}{m}$. By de Moivre, $\omega^m = 1$, and multiplying by ω rotates the plane by $360°/m$ about the origin.

Julia sets ($m = d$). From $\omega^d = 1$ it follows that $f_{d,c}(\omega z) = \omega^d z^d + c = f_{d,c}(z)$. The points z and ωz therefore have the same image after just one step, and either both remain bounded or both escape. The filled Julia set is therefore invariant under rotation by $360°/d$. For $d = 2$ this is the point symmetry $z \mapsto -z$ from Chapter 5.

Multibrot sets ($m = d-1$). Now let $\omega^{d-1} = 1$, so $\omega^d = \omega$. We compare the critical orbit $z_0 = 0$, $z_{n+1} = z_n^d + c$ with the orbit $w_0 = 0$, $w_{n+1} = w_n^d + \omega c$ for the rotated parameter ωc. If $w_n = \omega z_n$ (clear for $n = 0$), then

$$w_{n+1} = (\omega z_n)^d + \omega c = \omega^d z_n^d + \omega c = \omega\,(z_n^d + c) = \omega z_{n+1}.$$

The orbit for the parameter ωc is therefore the orbit for the parameter c rotated by $360°/(d-1)$; both have the same moduli. Hence $c \in \mathcal{M}_d \Leftrightarrow \omega c \in \mathcal{M}_d$. Here one factor ω is "used up" in turning ω^d back into ω; this requires $\omega^{d-1} = 1$. For $d = 2$, $\omega = 1$: the Mandelbrot set has no rotational symmetry, only the reflection symmetry in the real axis, which holds for all $\mathcal{M}_d$ because $\overline{z^d + c} = \overline{z}^d + \overline{c}$.

8.2 General polynomials and rational functions

Several critical points. A polynomial of degree d has, counted with multiplicity, $d-1$ critical points, the zeros of its derivative. For $z^d + c$ they all coincide at 0; in general they have different orbits. Already for degree 3 one therefore needs two complex parameters, for instance in the normal form

$$P_{a,b}(z) = z^3 - 3a^2 z + b$$

with the critical points $\pm a$. The set of all (a,b) for which both critical orbits remain bounded is the cubic counterpart of $\mathcal{M}$. It lies in $\mathbb{C}^2$, so it has four real dimensions and can only be drawn in slices. Branner and Hubbard showed that it is compact and connected [37]. The cautious remark in the original about higher-dimensional parameter spaces (Chapter 3) finds its precise form here.

Why the dichotomy can fail. For all polynomials of degree $d \geq 2$ only a weakened dichotomy holds [10]: if *all* critical orbits remain bounded, the filled Julia set is connected; if *at least one* escapes, it falls apart into uncountably many pieces. For $z^2 + c$, in the second case every piece is a single point. For cubic polynomials, however, one critical orbit can escape while the other remains bounded. Then some of the infinitely many pieces can be shrunken, distorted copies of a connected quadratic Julia set: the Julia set is not connected, but it is not a Cantor dust either [38]. Exactly which polynomials have Cantor sets as Julia sets is stated by a conjecture of Branner and Hubbard that was proved in 2009 [39]. That a quadratic Julia set can be "hidden" inside a cubic polynomial is explained by Douady and Hubbard's theory of polynomial-like mappings [19]; it is also the reason for the small Apfelmännchen from Section 3.5.

Rational functions and Newton's method. Julia and Fatou developed their theory from the outset for rational functions $R(z) = P(z)/Q(z)$. An example from the school syllabus is

Newton's method: for a polynomial p one iterates the rational function

$$N_p(z) = z - \frac{p(z)}{p'(z)}.$$

Every simple zero of p is a strongly attracting fixed point of N_p. For $p(z) = z^3 - 1$ there are three zeros, 1 and $-\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$, and hence three basins of attraction. Their common boundary is the Julia set of N_p, a fractal: at each of its points all three basins meet. If one starts near this boundary, a tiny change in the starting value decides which zero one finds [4]. For rational functions the Fatou components can be completely classified; the decisive step was Sullivan's proof in 1985 that there are no "wandering" components [40].

8.3 The bridge to transcendental functions

Euler's formula from the polar form. In Section 1.6.2 we saw that for $E(\varphi) = \cos\varphi + i\sin\varphi$ the addition formulae give

$$E(\varphi_1)\,E(\varphi_2) = E(\varphi_1 + \varphi_2), \qquad E(0) = 1.$$

This is the same rule as for powers, $a^x a^y = a^{x+y}$. The derivative fits too: $E'(\varphi) = -\sin\varphi + i\cos\varphi = i\,E(\varphi)$. So E behaves as $e^{i\varphi}$ would have to behave if the rule $(e^{k\varphi})' = k\,e^{k\varphi}$ is also to hold for $k = i$. One therefore defines

$$e^{i\varphi} = \cos\varphi + i\sin\varphi \qquad \text{(Euler's formula)}.$$

The polar form thus becomes $z = r\,e^{i\varphi}$, and the rule "multiply the moduli, add the angles" is simply the law of exponents. For $\varphi = \pi$ one obtains $e^{i\pi} = -1$. The same formula results from substituting $i\varphi$ into the power series of the exponential function; that is the usual proof.

The complex exponential function. For $z = x + iy$ one sets

$$e^z = e^x e^{iy} = e^x(\cos y + i\sin y).$$

So $|e^z| = e^x$ and $\arg e^z = y$: the real part determines the modulus, the imaginary part the angle. In particular $e^z \neq 0$, and $e^{z+2\pi i} = e^z$: the exponential function has the purely imaginary period $2\pi i$, takes every value $w \neq 0$ infinitely often, and its images repeat in the vertical direction every $2\pi \approx 6.28$ units. From e^{iz} and e^{-iz} one obtains

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}, \qquad \sin z = \frac{e^{iz} - e^{-iz}}{2i}.$$

For real z these are the familiar functions with period 2π. In the complex plane, however, they are unbounded: for example, $\cos(i) = \frac{e^{-1}+e}{2} \approx 1.543$, and $|\sin(x+iy)|$ grows roughly like $\frac{1}{2}e^{|y|}$.

Why "escaping" now means something different. For a polynomial of degree $d \geq 2$, $|P(z)|$ is much larger still when $|z|$ is large; the escape radius 2 of the 1990 algorithm rests on this. For e^z this no longer holds: $z = -1000$ has a huge modulus, but e^{-1000} lies extremely close to 0. One says that e^z has an *essential singularity* at ∞: outside every circle, however large, e^z takes every value except 0 infinitely often. There is no escape radius. The same holds for $\sin z$ and $\cos z$; for example, they remain bounded on the whole real axis.

A second consequence is even more surprising: for λe^z, $\lambda\sin z$ and $\lambda\cos z$, all points whose orbit tends to ∞ lie in the *Julia set* [41]. For $z^2 + c$ the escaping points formed an open region of the Fatou set outside K_c. Now they are used to draw the Julia set itself. This must be borne in mind when looking at the following images.

8.4 The exponential family λe^z

We iterate $E_\lambda(z) = \lambda e^z$ with $\lambda \neq 0$. Since $E'_\lambda = E_\lambda$ is never zero, there is no critical point. Its role is taken over by the value 0, which E_λ never takes but which the function approaches arbitrarily closely in the left half-plane. As for $z^2 + c$, then, the orbit of a single "special" value, $0, \lambda, \lambda e^\lambda, \ldots$, governs the dynamics [42].

Two fixed points and their merging. For real $\lambda > 0$ the graphs of $y = \lambda e^x$ and $y = x$ intersect twice for small λ and not at all for large λ. In the limiting case they touch: $\lambda e^x = x$ and at the same time $\lambda e^x = 1$ (equal gradient). From this it follows that $x = 1$ and $\lambda = 1/e \approx 0.3679$. For $0 < \lambda < 1/e$ the smaller fixed point q is attracting, since $E'_\lambda(q) = \lambda e^q = q < 1$; for $\lambda = 0.3$, $q \approx 0.4894$. The orbit of 0 tends to q, and the Fatou set is precisely the basin of attraction of q [43].

The Cantor bouquet. The Julia set then consists of uncountably many pairwise disjoint curves that run off to infinity to the right, each with one endpoint. They are called *hairs*, and the whole is called a *Cantor bouquet* [44, 45]. The hairs lie so densely side by side that each is approximated arbitrarily closely from both sides by other hairs, much as every point of the Cantor set is a limit of other points. On the hairs the orbits flee to the right; the points in between are drawn towards q.

Here the Hausdorff dimension shows an astonishing side: the Julia set has dimension 2 [33], but area 0. The hairs without their endpoints have dimension only 1, as one expects of curves. The full dimension therefore resides in the set of endpoints [46, 47].

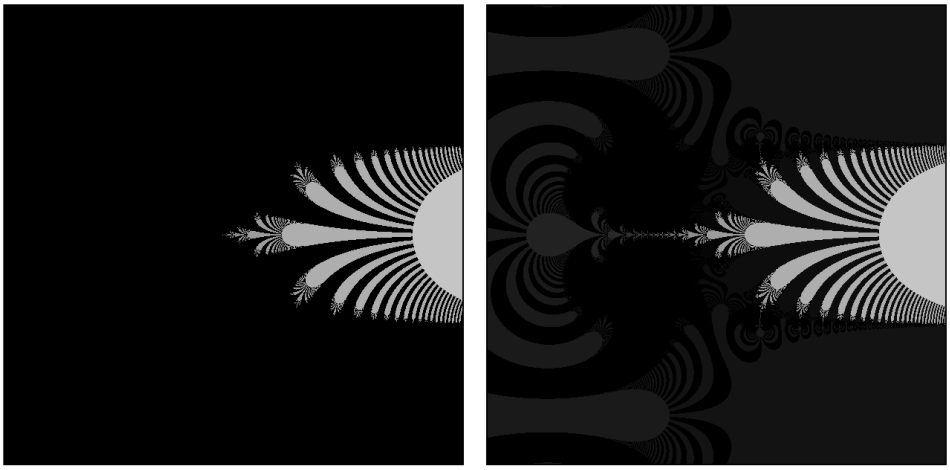


Figure 9: The exponential family λe^z for $\lambda = 0.3$ (left) and $\lambda = 0.4$ (right). *Note: the shades of grey are reversed compared with the images of* $z^2 + c$. Light points are those whose orbit flees to infinity to the right; they lie on the hairs and belong to the Julia set, not to the Fatou set. On the left, black is the basin of attraction of the attracting fixed point $q \approx 0.49$, that is, the Fatou set. On the right there is no longer an attracting fixed point; the whole plane is Julia set, and dark points have merely not yet escaped within the iteration limit.

The explosion at $\lambda = 1/e$. At $\lambda = 1/e$ the two fixed points merge at the point 1. For real

$\lambda > 1/e$ there is no longer a real fixed point, the orbit of 0 travels along the real axis to infinity, and the Julia set is the whole plane [48]. A small step from $\lambda = 0.36$ to $\lambda = 0.37$ thus turns a nowhere dense set into the whole of $\mathbb{C}$: an *explosion* of the Julia set. For $z^2 + c$ there is nothing comparable, because there the Julia set is always bounded and has no interior. For $\lambda = 1$, that is, for e^z itself, the orbit $0 \to 1 \to e \to e^e \to \cdots$ likewise runs off to infinity, and $J(e^z)$ is also the whole plane: in every neighbourhood of every point, however small, there are starting values with completely different behaviour.

Incidentally, the calculation explains the power tower a, a^a, $a^{(a^a)}$, ... for $a > 1$: with $\lambda = \ln a$ and $w = \lambda t$, the step $t \mapsto a^t$ becomes the mapping $w \mapsto \lambda e^w$. The tower therefore converges exactly for $\ln a \le 1/e$, that is, for $a \le e^{1/e} \approx 1.4447$ [42].

> **Historical context** Pierre Fatou began iterating entire transcendental functions in 1926 and asked whether the Julia set of e^z can be the whole plane. Michał Misiurewicz proved this in 1981 [49]. With today's theory the reason can be put into words: every kind of Fatou component that is possible at all for e^z would have to "hold on to" the orbit of the special value 0; but this orbit escapes [42]. The images of the Cantor bouquet and of the explosion come from work by Robert Devaney and co-authors from 1984 onwards [44, 48].

8.5 Sine and cosine

Sine and cosine have critical points, at $\frac{\pi}{2} + k\pi$ for $\sin z$ and at $k\pi$ for $\cos z$, but only the critical values ± 1. So once again there are only a few "special" orbits.

- $f(z) = \sin z$: the fixed point 0 has $\sin'(0) = \cos 0 = 1$. It attracts real starting values only very slowly, much like the fixed point $\frac{1}{2}$ for the cauliflower $c = 0.25$ from Chapter 5. The Fatou set of $\sin z$ consists of points that tend to 0 in this way [42].
- $f(z) = \cos z$: anyone who presses the cosine key on a calculator (in radians) again and again ends up at $x^* \approx 0.739085$, the only real fixed point of $\cos x = x$. Since $|\cos'(x^*)| = \sin x^* \approx 0.674 < 1$, it is attracting. Both critical values ± 1 are attracted by it, and the Fatou set is its basin of attraction.

Figure 10 again shows hairs, this time pointing up and down, because sine and cosine can become large only in the vertical direction. The light regions, however, look not like thin curves but like areas. This is not merely an illusion caused by the image resolution: McMullen proved that the Julia set of $\sin(\alpha z + \beta)$ has *positive area* for all $\alpha \neq 0$ and all β [33]. Since $\cos z = \sin(z + \frac{\pi}{2})$, this also holds for $\cos z$, even though the dynamics there, with an attracting fixed point, are as "calm" as can be. For $z^2 + c$ with an attracting fixed point, by contrast, the Julia set has area zero.

8.6 The tangent: a "fractal" that is a straight line

The tangent $\tan z = \sin z / \cos z$ has poles at $\frac{\pi}{2} + k\pi$; it is not entire but *meromorphic*. It has no critical points; its special values are $\pm i$: as $\operatorname{Im} z \to \pm\infty$, $\tan z$ tends to $\pm i$ without ever reaching these values. We consider $\lambda \tan z$ with real $\lambda > 0$. A calculation with the formulae from Section 8.3 shows that $\operatorname{Im} \tan z$ has the same sign as $\operatorname{Im} z$: the upper half-plane is mapped into itself, and so is the lower one. In these half-planes things proceed in an orderly way; chaos can only take place on the real axis. Devaney and Keen, and Keen and Kotus, described what happens there [50, 51]:

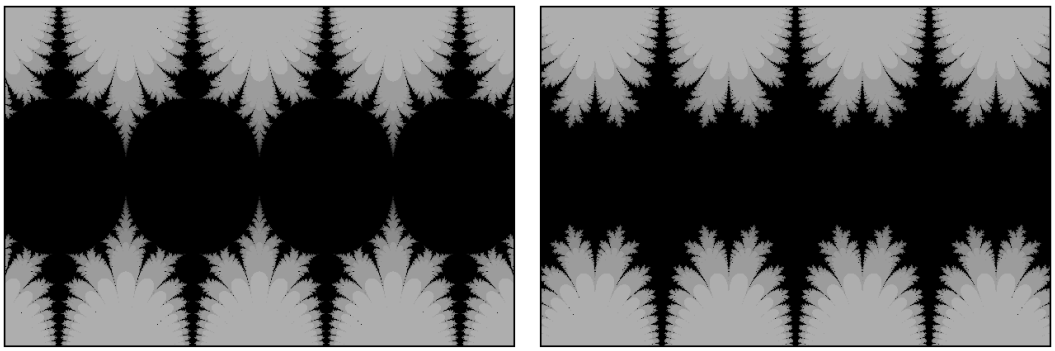


Figure 10: Escape-time images for $f(z) = \sin z$ (left) and $f(z) = \cos z$ (right). Light: $|\mathrm{Im}\, z|$ becomes large and the orbit escapes (the lighter, the faster). Black: the orbit remains bounded within the iteration limit. As for λe^z, the escaping, light points belong to the Julia set. The patterns repeat horizontally with period 2π.

- For $0 < \lambda < 1$, 0 is an attracting fixed point ($(\lambda \tan)'(0) = \lambda < 1$). The Julia set lies on the real axis and is locally a Cantor set there. The Fatou set is a single connected region.
- For $\lambda > 1$ the Julia set is the whole real axis. There are two attracting fixed points $\pm i\xi_0$ with $\xi_0 = \lambda \tanh \xi_0$, for $\lambda = 2$ roughly $\xi_0 \approx 1.915$; the upper and lower half-planes are their basins of attraction.

In the limiting case $\lambda = 1$, 0 is a parabolic fixed point; here too the literature gives $J = \mathbb{R}$.

For $\lambda > 1$ the Julia set is therefore a straight line, the smoothest of all shapes, with Hausdorff dimension 1. How does that square with "fractal"? The Julia set is defined by the *dynamics*: on it, small changes in the starting value lead to completely different orbits; near the poles the tangent flings neighbouring points arbitrarily far apart. Whether the set looks geometrically rough is a separate question. You already know a similar example: the line segment $J_{-2} = [-2, 2]$ (Chapter 6). The case $0 < \lambda < 1$, on the other hand, corresponds to the case $c < -2$: a Cantor set on the real axis.

8.7 The logarithm: an orbit as a spiral

The logarithm is meant to solve $e^w = z$ for w. With $z = r\, e^{i\varphi}$ and $w = u + iv$ this means $e^u = r$ and $e^{iv} = e^{i\varphi}$, so

$$w = \ln r + i(\varphi + 2\pi k), \qquad k \in \mathbb{Z}.$$

Because of the period $2\pi i$ of e^z, every $z \neq 0$ has infinitely many logarithms; the complex logarithm is *multivalued*. For iterating, one chooses the *principal branch* at every step:

$$\mathrm{Log}\, z = \ln|z| + i\, \mathrm{Arg}\, z, \qquad -\pi < \mathrm{Arg}\, z \leq \pi.$$

It is undefined at $z = 0$ and jumps along the negative real axis; the theory of this chapter therefore cannot be applied directly. An orbit can even break off: since $\mathrm{Log}\, 1 = 0$, for $z_0 = 1$ the second step is already undefined.

Nevertheless, the iteration behaves very tidily. Figure 11 shows the orbit of $z_0 = 2 + 0.5i$:

$$z_1 \approx 0.723{+}0.245i, \quad z_2 \approx -0.269{+}0.327i, \quad z_3 \approx -0.860{+}2.261i, \quad z_4 \approx 0.883{+}1.934i, \quad \ldots$$

It winds in a spiral towards the point

$$p \approx 0.3181 + 1.3372i,$$

which satisfies $\operatorname{Log} p = p$, that is, $e^p = p$. It cannot get away: the principal branch maps the upper half-plane into itself, because $0 < \operatorname{Arg} z < \pi$ holds there, and by a classical theorem of complex analysis (the Denjoy–Wolff theorem) every orbit in the upper half-plane then converges to the only fixed point there. Orbits in the lower half-plane correspondingly tend to $\overline{p}$.

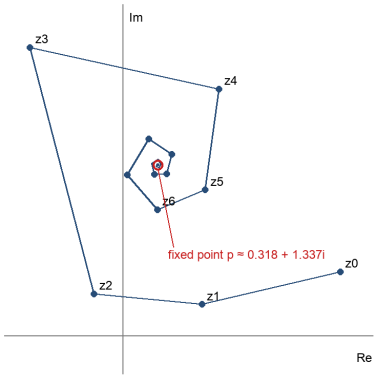


Figure 11: The orbit of the principal branch of the logarithm starting from $z_0 = 2 + 0.5i$. After a few steps it spirals into the fixed point $p \approx 0.318 + 1.337i$.

Why a spiral? Close to p, Log acts like multiplication by the derivative $\operatorname{Log}'(p) = 1/p$: $z_{n+1} - p \approx \frac{1}{p}(z_n - p)$. By Section 1.6.2 this means that the distance to p shrinks at each step by the factor $1/|p| \approx 0.7275$, and the connecting vector rotates by $-\arg p \approx -76.6°$. The result is a clockwise spiral with about 4.7 steps per revolution.

For e^z, p is also a fixed point, but there it is repelling, since $|(e^z)'(p)| = |e^p| = |p| \approx 1.375 > 1$. This fits with $J(e^z) = \mathbb{C}$, because repelling fixed points lie in the Julia set. Iterating the logarithm means iterating e^z backwards, and backwards, repelling points become attracting. The same principle is used to draw Julia sets by backward iteration.

8.8 Outlook: Apfelmännchen in unexpected places

The Mandelbrot set is more than the picture of a single formula. McMullen showed that in every family of *rational* functions depending holomorphically on one complex parameter, the set of parameters at which the behaviour changes qualitatively is either empty or contains (distorted) copies of the boundary of a Multibrot set $\mathcal{M}_d$; small Mandelbrot sets even lie densely there [52]. This is why, for example, Apfelmännchen are found in the parameter plane of Newton's method for cubic polynomials [19]. The theorem does not hold for the transcendental families of this chapter. Their parameter planes are studied in their own right: for λe^z, the main cardioid corresponds to the set of λ with an attracting fixed point, whose boundary passes through the explosion point $\lambda = 1/e$; in addition there are infinitely many regions with attracting cycles

and, between them, “parameter hairs” [42]. The set at which the behaviour changes is, like $\mathcal{M}$, connected [53].

Try it yourself

1. Check Euler’s formula. In Python, `1j` stands for i. Compare `cmath.exp(1j*phi)` with `math.cos(phi) + 1j*math.sin(phi)` for a few angles and check the period:

```
import cmath, math
z = 0.7 - 1.2j
print(cmath.exp(z), cmath.exp(z + 2j*math.pi))   # equal!
print(cmath.exp(z + 1j*math.pi))          # not equal: sign
```

Why is πi not a period, even though $e^{z+\pi i}$ has the same modulus as e^z?

2. The logarithm spiral. Iterate the principal branch `cmath.log` starting from $2 + 0.5i$ and watch the distance to the fixed point p:

```
import cmath
p = 0.3181315052 + 1.3372357014j
z = 2 + 0.5j
for n in range(1, 21):
    z = cmath.log(z)
    print(n, z, abs(z - p))
print(cmath.exp(p) - p)                   # p is a fixed point of exp
```

Form the quotient of successive distances. Which number does it tend to? Compare it with $1/|p|$. What happens if you start at $2 - 0.5i$?

3. Symmetry by hand. For $d = 3$, calculate the orbit terms 0, c, $c^3 + c$ for $c = 0.3 + 0.2i$ and for $c = -0.3 - 0.2i$. Why is the second orbit the reflection of the first through the origin (Section 8.1)? Which parameter gives an orbit rotated by $120°$ when $d = 4$?

9 Open questions

What you already know

In Chapter 3 the Mandelbrot set $\mathcal{M}$ was introduced as the set of parameters c for which the orbit of $z_0 = 0$ under $f_c(z) = z^2 + c$ remains bounded. Chapter 5 set the Julia sets J_c in the dynamical plane alongside it, and Chapter 6 showed that K_c is either connected or Cantor dust. With the Hausdorff dimension from Chapter 7 one can measure the "size" of sets that neither length nor area describes sensibly, and Chapter 8 went beyond $z^2 + c$ to $z^d + c$, λe^z and other functions. This chapter shows where established knowledge about these objects ends today.

Mathematics is not finished. The new edition of the 1990 paper already shows this: the "Historical context" box in Chapter 4 points to a statement that had not yet been proved in 1990. Only some years later was it shown that the boundary of the Mandelbrot set has Hausdorff dimension 2 [31]. A picture that appeared on the screen in 1990 thus shows structures about which research still does not know everything today.

We proceed in the same way for each open question: first it is described in simple terms, then the precise formulation follows, and finally the *status as of October 2026*, with a source. One distinction matters here: in mathematics a result counts as established only once its proof has been checked by experts and published in a journal (refereeing, known as *peer review*). Papers that so far exist only as a *preprint* on the arXiv server are explicitly identified as such in what follows.

9.1 Open questions about the Mandelbrot set and Julia sets

Is the Mandelbrot set locally connected? (MLC)

In simple terms. In 1982 Douady and Hubbard proved that $\mathcal{M}$ is connected [16]: the set consists of a single piece, even though the many small "Apfelmännchen" at the edge look like islands in the pictures. Being *locally* connected is a stronger property. It means: anyone standing at a point of $\mathcal{M}$ can move about within a small neighbourhood inside $\mathcal{M}$ without having to make a large detour. The abbreviation MLC stands for "Mandelbrot set is locally connected".

Precise formulation.

For every $c \in \mathcal{M}$ and every neighbourhood U of c there is a neighbourhood $V \subseteq U$ of c such that $V \cap \mathcal{M}$ is connected.

Why it matters. If MLC were true, the picture of $\mathcal{M}$ could be described completely: Douady, Hubbard and Thurston built a combinatorial model of $\mathcal{M}$, a disc in which certain points are "pinched together" according to fixed rules. MLC is precisely the statement that this model correctly reproduces the actual Mandelbrot set [10]. Moreover, MLC implies the density of hyperbolicity (next section).

Status 2026: open. Around 1990 Yoccoz showed that $\mathcal{M}$ is locally connected at all parameters that are not "infinitely renormalisable"; roughly speaking, these are all points that do not lie in infinitely many nested small copies of $\mathcal{M}$ [54]. For many of the remaining parameters there are partial results, most recently by Dudko and Lyubich [55]. A survey article for the 2026 International Congress of Mathematicians still treats MLC as an open problem [56] (preprint).

Density of hyperbolicity

In simple terms. In the interior of $\mathcal{M}$ one sees the main cardioid and countless discs and bulbs (Section 3.5). For parameters c in these regions the orbit of 0 behaves particularly "tamely": it is attracted by a cycle, such as a fixed point or a 2-cycle. Such parameters are called *hyperbolic*. The question is whether this tame behaviour can be found everywhere nearby, or whether the interior of $\mathcal{M}$ also contains regions of an entirely different kind.

Precise formulation.

The set of $c \in \mathbb{C}$ for which f_c has an attracting cycle is dense in $\mathcal{M}$.

Equivalently: every component of the interior of $\mathcal{M}$ is hyperbolic.

Status 2026: real case solved, complex case open. For real parameters $c \in [-2, \frac{1}{4}]$, that is, for the functions $x \mapsto x^2 + c$ on the number line, Lyubich and, independently, Graczyk and Świątek proved density in 1997 [57, 58]. Kozlovski, Shen and van Strien extended this in 2007 to all real polynomials [59]. For complex c the question is open; MLC would answer it [10].

How large is the area of $\mathcal{M}$?

In simple terms. The Mandelbrot set lies in a rectangle of roughly 2.5×2.5 units. What exactly is its area? There is no known formula.

What is known. If one counts, on a very fine grid, the points whose orbit does not escape, one obtains

$$\text{area}(\mathcal{M}) \approx 1.50659.$$

This is a *numerical estimate* from very large computations, not refereed by experts and not a proof [60]. Only bounds have been proved. In 1993 Fisher and Hill stated in an unpublished manuscript [61]:

$$1.50296686 < \text{area}(\mathcal{M}) < 1.57012937.$$

Another approach uses a series expansion that leads to an exact formula for the area. Truncating the series gives upper bounds, but these approach the true value only very slowly [62].

Status 2026: open. An exact value is not known.

Does the boundary of $\mathcal{M}$ have area zero?

In simple terms. The boundary of a circle is a line and therefore has no area. The boundary of $\mathcal{M}$, by contrast, is so jagged that it has Hausdorff dimension 2 [31], and is thus in this sense "as large as a surface" (Chapter 7). Nevertheless, a set of dimension 2 can have area zero. What holds for $\partial\mathcal{M}$?

Precise formulation.

$$\text{Is} \quad \text{area}(\partial\mathcal{M}) = 0 \ ?$$

Status 2026: open [10]. The question also concerns pixel counting: if the boundary had positive area, there would be a "blurred" portion that no picture can cleanly assign to the interior or the exterior.

A surprisingly solved question: Julia sets of positive area

In simple terms. All the Julia sets J_c in the gallery in Chapter 5 are thin structures, curves or dust, without area. For a long time it was open whether *every* Julia set of $z^2 + c$ has area zero.

Precise formulation.

Is there a $c \in \mathbb{C}$ for which J_c has positive area?

Status: solved, the answer is yes. In 2012 Buff and Chéritat constructed quadratic polynomials whose Julia set has positive area [63]. In 2022 Avila and Lyubich found further examples, the so-called Feigenbaum Julia sets [64].

9.2 Open questions in fractal geometry

The following questions have nothing to do with $z^2 + c$, but they do have to do with the tools from Chapter 7: dimension, coverings and self-similarity.

The Kakeya problem

In simple terms. In 1917 Kakeya asked how little area one needs in order to turn a needle of length 1 in the plane once completely round. Besicovitch showed that there are sets that contain a line segment of length 1 in *every* direction and nevertheless have area (in space: volume) 0. Such sets are called Kakeya sets. The conjecture says: even though they have no volume, they are not "thin": their dimension is full.

Precise formulation.

Every compact set $K \subseteq \mathbb{R}^n$ that contains a line segment of length 1 in every direction has Hausdorff dimension n.

Status 2026: proved in the plane; in space a generally accepted proof (main article still a preprint); open from dimension 4 upwards. For $n = 2$ the statement has been proved since 1971 (Davies). For $n = 3$ Hong Wang and Joshua Zahl presented a proof in February 2025 [65]. The main article has so far appeared only as a preprint; important building blocks, however, have been published after refereeing [66], a simplified version of the proof has been accepted for publication [67], and experts consider the proof to be correct [68]. On 23 July 2026 Hong Wang received the Fields Medal, one of the highest distinctions in mathematics, in part for this work [69]. For $n \geq 4$ the conjecture is open.

Falconer's distance problem

In simple terms. Take a set of points E in the plane and write down all the distances between any two of its points. If E is "large enough", these distances should fill a whole stretch of the number line and not just be scattered values. In 1985 Falconer conjectured that a dimension greater than 1 is already sufficient.

Precise formulation.

$$E \subseteq \mathbb{R}^2 \text{ compact, } \dim_{\mathrm{H}} E > 1 \implies \Delta(E) = \{\, |x-y| : x,y \in E \,\} \text{ has positive length.}$$

In $\mathbb{R}^n$ the threshold is $n/2$ instead of 1.

Status 2026: open. In the plane it has been proved that $\dim_{\mathrm{H}} E > \frac{5}{4}$ suffices (Guth, Iosevich, Ou and Wang, 2020) [70]. The gap between 1 and $\frac{5}{4}$ has not yet been closed.

The exact overlaps conjecture

In simple terms. The Cantor set consists of two reduced copies of itself that do not overlap; its dimension therefore follows from the formula in Section 4.2. If the copies overlap, the dimension can be smaller. The conjecture says: on the number line the dimension becomes smaller *only if* two copies coincide exactly. Chance near-overlaps do no harm.

Precise formulation. Let $S_i(x) = r_i x + t_i$ with $0 < |r_i| < 1$ be finitely many similarity maps with the self-similar set F and the similarity dimension s, that is, $\sum_i |r_i|^s = 1$. If no two distinct compositions $S_{i_1} \circ \cdots \circ S_{i_k}$ and $S_{j_1} \circ \cdots \circ S_{j_l}$ are equal, then

$$\dim_{\mathrm{H}} F = \min\{1, s\}.$$

Status 2026: on the number line a proof claimed, not yet checked. Hochman proved the statement in 2014 under an additional separation condition [71], and Rapaport in 2022 for algebraic contraction factors [72]. On 7 October 2026, eleven days before the date of this text, Kittle and Kogler published a proof of the full one-dimensional case as a preprint, together with a computer check in the proof assistant Lean [73]. According to their own account, an AI system (GPT-6 Astra) found the first proof; an independent proof produced by OpenAI had become public one day earlier. The paper is *not refereed*. One should therefore say: a proof is claimed, but it is not yet established. In higher dimensions the question is open.

Furstenberg's $\times 2 \times 3$ conjecture

In simple terms. If one multiplies a number from $[0, 1)$ by 2 and discards the integer part, one shifts its binary representation by one place. Multiplying by 3 likewise shifts the representation in base 3. Furstenberg conjectured that the two operations are so "independent" that nothing interesting fits both at once.

Precise formulation.

Every ergodic probability measure on $[0, 1)$ *that is invariant under* $x \mapsto 2x \bmod 1$ *and under* $x \mapsto 3x \bmod 1$ *is either the ordinary length measure or is supported on finitely many rational numbers.*

Status 2026: open; a related conjecture is solved. In 1990 Rudolph proved the case of "positive entropy" [74]; the rest is open. A related conjecture of Furstenberg on the dimension of intersections of $\times 2$- and $\times 3$-invariant sets was proved in 2019 by Shmerkin and, independently, by Wu [75, 76].

Question	since	Status October 2026
Local connectivity of $\mathcal{M}$ (MLC)	1980s	open
Density of hyperbolicity	1920s	real case solved (1997), complex open
Area of $\mathcal{M}$	c. 1990	open, only bounds
Area of $\partial\mathcal{M}$ equal to 0?	1980s	open
Julia sets of positive area	1980s	solved: yes (2012)
Kakeya conjecture	1917	$n = 2$ solved, $n = 3$ proved (preprint), $n \geq 4$ open
Falconer's distance problem	1985	open ($\frac{5}{4}$ reached)
Exact overlaps	2014	$\mathbb{R}$: proof claimed (preprint)
Furstenberg's $\times 2 \times 3$ conjecture	1967	open

Table 2: The open questions of this chapter at a glance. The column "since" indicates only approximately since when the question has been posed in its present form.

9.3 How conjectures end

A conjecture can end in two ways: it is proved, or someone finds a counterexample. Fractal geometry offers a fine example of each.

Proved: the frontier of Brownian motion

In simple terms. A particle that jumps randomly back and forth in the plane traces out a tangled path (Brownian motion). If one colours only the outer boundary of this path, one obtains a jagged curve. On the basis of computer pictures, Mandelbrot conjectured that this boundary has dimension $\frac{4}{3}$.

Precise formulation. With probability 1, for the path $B[0,1]$ of a planar Brownian motion,

$$\dim_{\mathrm{H}}\big(\text{outer boundary of } B[0,1]\big) = \tfrac{4}{3}.$$

Result. Lawler, Schramm and Werner proved this in 2001, using a method that was new at the time, the Schramm–Loewner evolution [36]. This work was among the reasons for Werner's Fields Medal in 2006. Here, then, a conjecture that had arisen from looking at pictures turned out to be correct.

Refuted: Eremenko's conjecture

In simple terms. In Chapter 8 functions such as λe^z were studied. For them there is the *escaping set* $I(f)$: the points whose orbit runs off to infinity. In the pictures it looks like hairs that reach all the way to infinity. In 1989 Eremenko asked whether this is always so: does every connected piece of $I(f)$ reach to infinity [77]?

Precise formulation.

For every transcendental entire function f, every connected component of $I(f) = \{z : f^n(z) \to \infty\}$ is unbounded.

Result. A stronger version (every escaping point can be joined to ∞ by a curve in $I(f)$) was refuted in 2011 by Rottenfußer, Rückert, Rempe and Schleicher [78]. The conjecture itself was refuted by Martí-Pete, Rempe and Waterman; their paper appeared in 2025 in the *Journal of the American Mathematical Society* [79]. They construct an entire function whose escaping set has a bounded piece. In an explanatory note (preprint, September 2026) they describe how such a piece can even be a single point [80]. Part of the original idea remains true: Rippon and Stallard had shown in 2005 that $I(f)$ always has *at least one* unbounded piece [81].

This example shows why pictures alone prove nothing. For the exponential functions λe^z from Chapter 8 even the stronger version holds [47]; there the hairs in the pictures do not deceive. The counterexamples, by contrast, are specially constructed, very particular functions. A conjecture that holds for all known examples can nevertheless be false.

9.4 What you can explore yourself today

Research is not only about proving. Many of the questions in this chapter arose from experiments on the computer, just like the pictures in the 1990 paper. Today one can use a short program to produce in seconds pictures that required long computing times in 1990, recompute numerical values oneself and follow research as it happens: new results usually appear first, freely accessible, on `https://arxiv.org`, articles on complex dynamics for instance in the category *Dynamical Systems* (math.DS). Popular-science magazines publish readily understandable reports on new proofs, for example *Quanta Magazine* (`https://www.quantamagazine.org`).

Anyone reading there should keep in mind the distinction made in this chapter: a preprint is a *proposed* proof. Only careful checking by others turns it into established knowledge, and sometimes that takes years.

Try it yourself

1. **The area of $\mathcal{M}$ by pixel counting.** The following program lays a grid over the upper half of $\mathcal{M}$ and counts the grid points whose orbit has not escaped after 200 steps. Because of the symmetry about the real axis, the result is doubled.

```
N, maxit = 400, 200                      # grid points per axis, iterations
x0, x1, y0, y1 = -2.0, 0.5, 0.0, 1.25    # upper half suffices (symmetry)
dx, dy = (x1 - x0) / N, (y1 - y0) / N
hits = 0
for i in range(N):
    for j in range(N):
        c = complex(x0 + (i + 0.5) * dx, y0 + (j + 0.5) * dy)
        z = 0
        for n in range(maxit):
            z = z * z + c
            if abs(z) > 2:
                break
        else:
            hits += 1
print("Area approximately", 2 * hits * dx * dy)
```

   With these values you obtain about 1.527. Compare this with the estimate 1.50659 and the proved bounds from Section 9.1. Change `N` and `maxit`. Why does the result tend to be too high? (Hint: which points does the program count even though they do not belong to $\mathcal{M}$?)

2. **Reading a report.** Look in *Quanta Magazine* for an article on the Kakeya problem or on the Mandelbrot set. Note which statements are presented there as proved and which as conjectured,

and compare with Table 2.

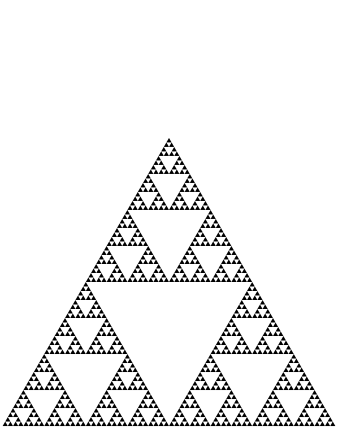
3. **Today's research.** On arXiv, open the list of new papers in the category math.DS (*Dynamical Systems*). Look for a title containing "Mandelbrot" or "Julia" and read the summary (*abstract*). Has the paper already appeared in a journal? If so, this is stated in the *Journal reference* field.

10 Appendix

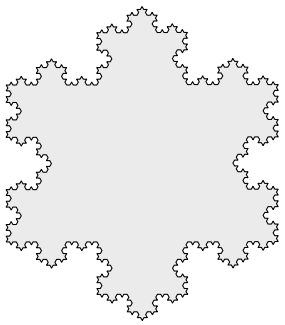
10.1 Various images

The following plates preserve the original selection of images and, where the original gives them, the coordinates and iteration depths. The Mandelbrot images with coordinates were recomputed in black and white with a Python program. The three-dimensional representation was likewise regenerated from the escape times. The colour plate was replaced by similar details, and the Koch curve and the scaling table were redrawn in LaTeX. The plate of IFS fractals shows three newly generated examples. The explanations beneath them make the mathematical meaning more precise.

Plate 1: Examples of iterated function systems (original page 31)



Sierpiński triangle



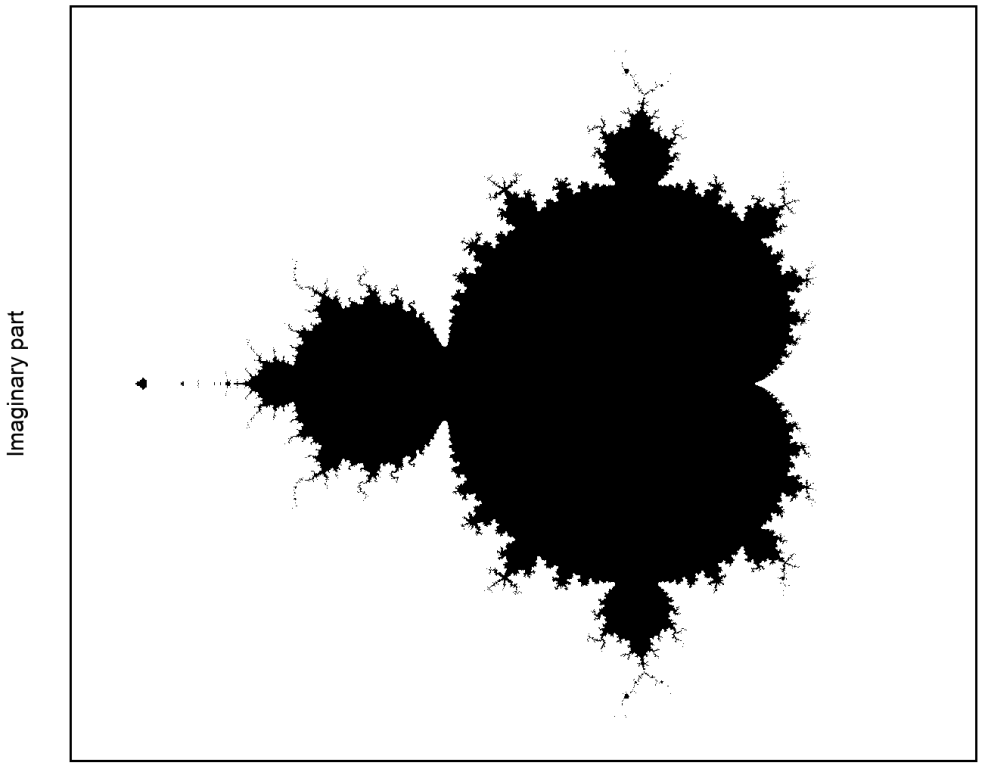
Koch snowflake



Barnsley fern

The original shows various IFS fractals, among them the Koch snowflake and the Sierpiński triangle. The remaining motifs lack labels or source references. The images were regenerated in 2026 and are restricted to examples with a known rule: the Sierpiński triangle and the snowflake are constructed exactly, and the fern is produced by the four affine maps given by Michael Barnsley (*Fractals Everywhere*, 1988). These figures are produced by iterated function systems. They do not belong to the Mandelbrot set, but they likewise show how complicated, self-similar structures can arise from a simple, repeated rule.

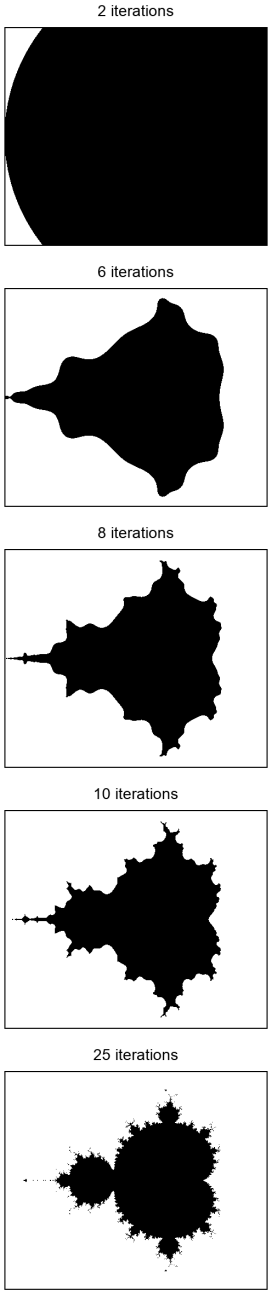
Plate 2: The Mandelbrot set in the complex plane (original page 32)




```
   Real part:  Minimum -2.00000000   Maximum 1.00000000
Imaginary part:  Minimum -1.25000000   Maximum 1.25000000
                   Iteration depth: 25
```

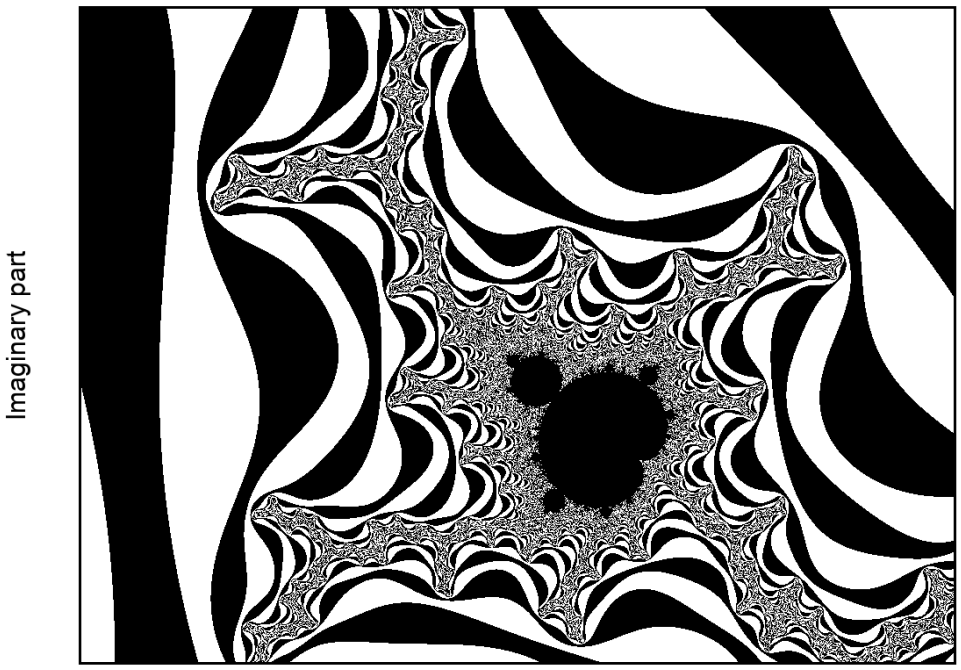
The black points have not escaped after 25 iterations. The picture is therefore a numerical approximation of the Mandelbrot set. A black point need not ultimately belong to the Mandelbrot set; its sequence may cross the escape boundary only after more than 25 steps. Convergence to zero is not assumed.

Plate 3: Effect of the iteration limit (original page 33)



The sequence of images shows the approximation after 2, 6, 8, 10 and 25 iterations. As the iteration limit rises, more and more points lying outside are recognised as escaping. The Mandelbrot set itself does not change; only the outer approximation becomes more accurate. Ever finer details therefore become visible, especially at the boundary, as the iteration depth increases.

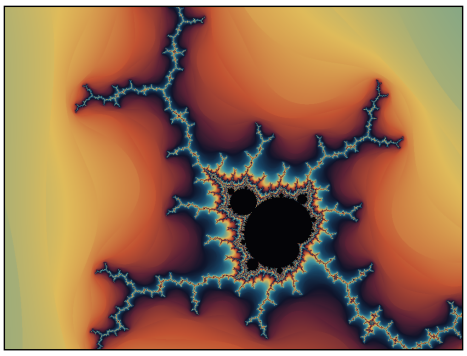
Plate 4: Magnification of a boundary region (original page 34)



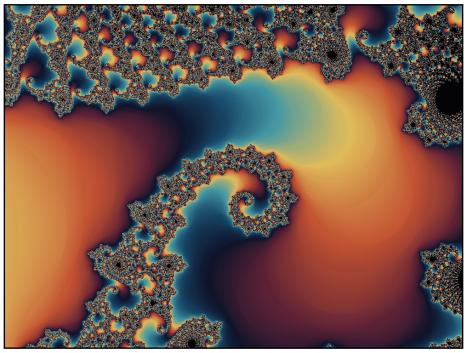

```
  Real part:  Minimum -0.19920000   Maximum -0.12954000
Imaginary part:  Minimum 1.01480000   Maximum 1.06707000
                 Iteration depth: 200
```

The image shows a magnification at the boundary of the Mandelbrot set. Further magnification brings out ever new structures and miniature copies. The region shown is only 0.06966 wide in the real part. The black and white bands denote different escape times; they are not additional parts of the Mandelbrot set.

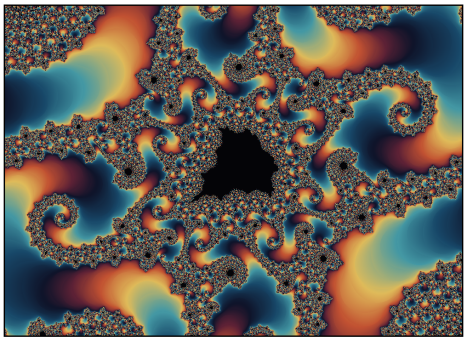
Plate 5: Coloured fractal images (original page 35)



Real part: Minimum −0.19920000 Maximum −0.12954000
Imaginary part: Minimum 1.01480000 Maximum 1.06707000
Iteration depth: 260



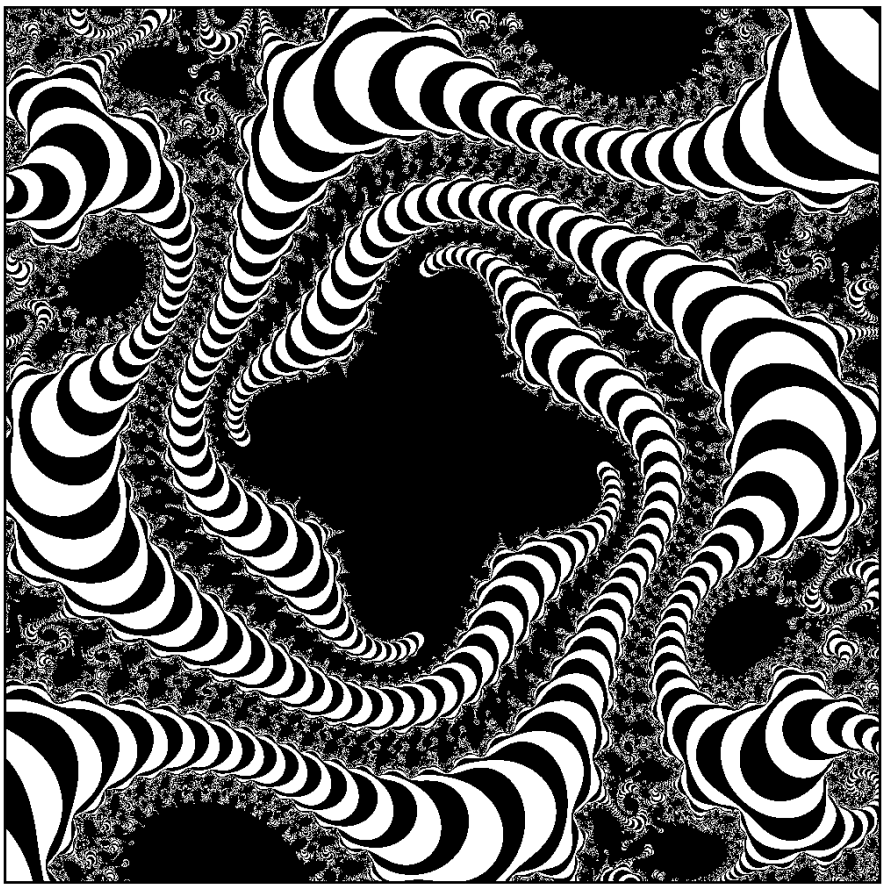
Real part: Minimum −0.74700000 Maximum −0.74000000
Imaginary part: Minimum 0.12900000 Maximum 0.13425000
Iteration depth: 320



Real part: Minimum −0.74543560 Maximum −0.74542150
Imaginary part: Minimum 0.11300370 Maximum 0.11301390
Iteration depth: 450

The original colour sheet survives in scan `IMG_1960` but gives no coordinates. For this reading edition, three similar details were therefore recomputed with a Python program; no claim is made to an exact reconstruction of the original images. The coordinate range used and the iteration depth are given beneath each image. The colours represent groups of similar escape times. They are not part of the Mandelbrot set as a mathematical set of points. A different colour assignment would therefore change the appearance, but not the computed sequences.

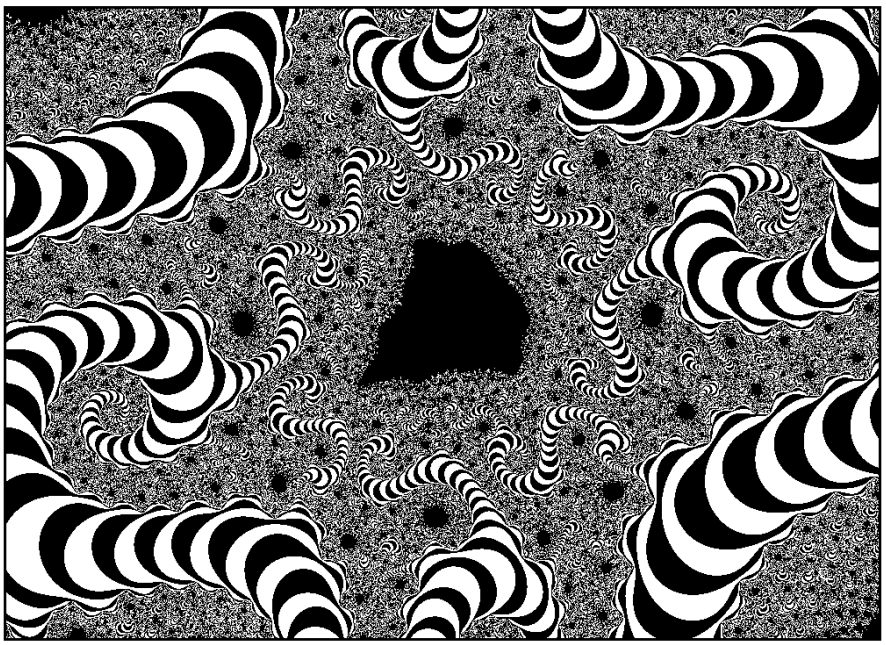
Plate 6: Black-and-white recomputation (original page 36)




```
   Real part:  Minimum -0.74638270   Maximum -0.74638210
Imaginary part:  Minimum -0.09850600   Maximum -0.09850540
                 Iteration depth: 350
```

A strong magnification with 350 iterations. The narrow black connections and recurring miniature copies illustrate the fine structure of the boundary. The width of the detail in the real part is only 0.00000060. The repetitions are similar to the main figure but not identical to it, since new structures appear at every level of magnification.

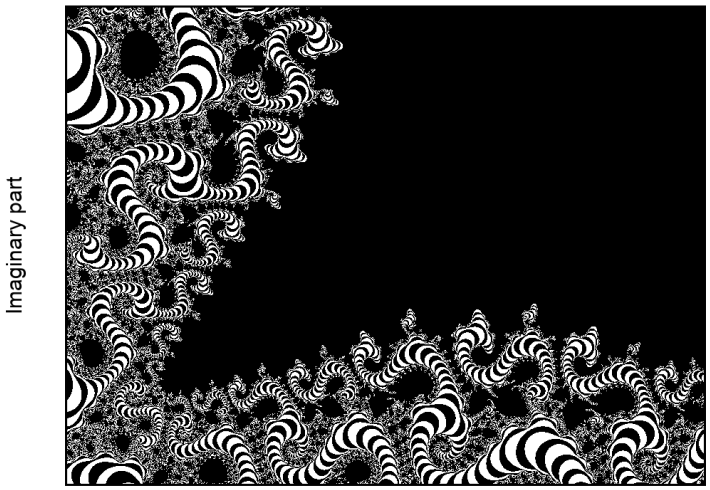
Plate 7: Another miniature copy (original page 37)



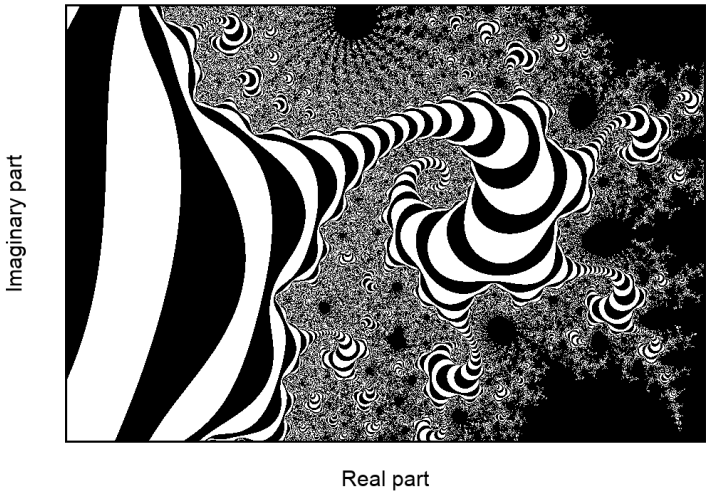

```
  Real part:  Minimum -0.74543560   Maximum -0.74542150
Imaginary part:  Minimum 0.11300370   Maximum 0.11301390
                 Iteration depth: 350
```

This detail, too, shows an approximate symmetry and a point-like miniature copy of the main figure. Its width in the real part is 0.00001410. The Mandelbrot set as a whole is exactly symmetric about the real axis. The local symmetry visible here, by contrast, is only approximate.

Plate 8: Two magnified details (original page 38)



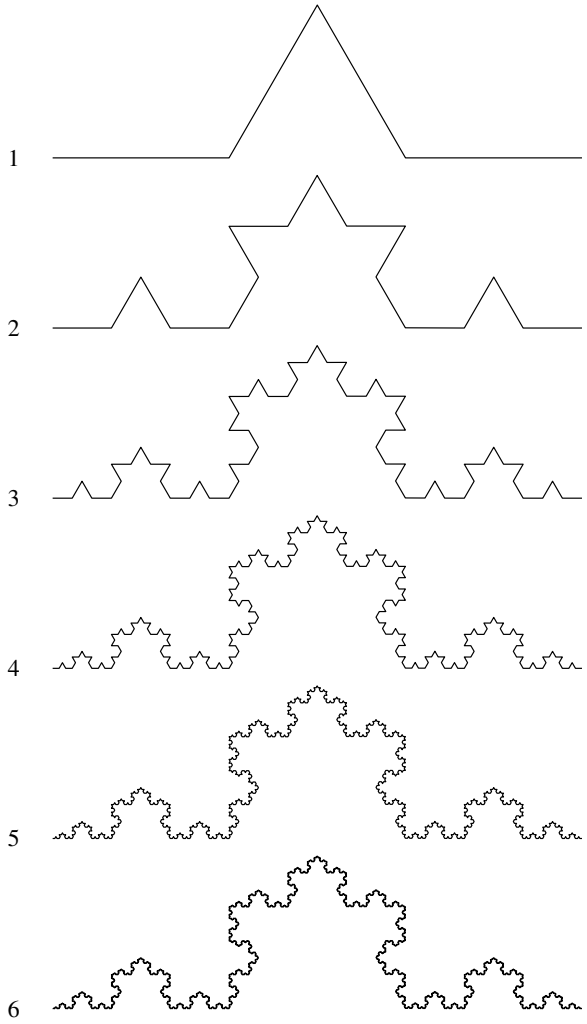
Real part: Minimum −0.74543010 Maximum −0.74542890
Imaginary part: Minimum 0.11300760 Maximum 0.11300850
Iteration depth: 350

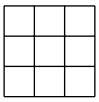
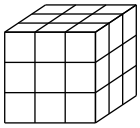


Real part: Minimum −0.75104000 Maximum −0.74080000
Imaginary part: Minimum 0.10835000 Maximum 0.11536000
Iteration depth: 300

The images document different details and iteration depths of the computer representation of the time. The upper detail spans only 0.0000012 in the real part, the lower one 0.01024. The bands arise from showing the escape times alternately in black and white. They make the speed of escape visible, but they do not prove that an individual point belongs to the Mandelbrot set.

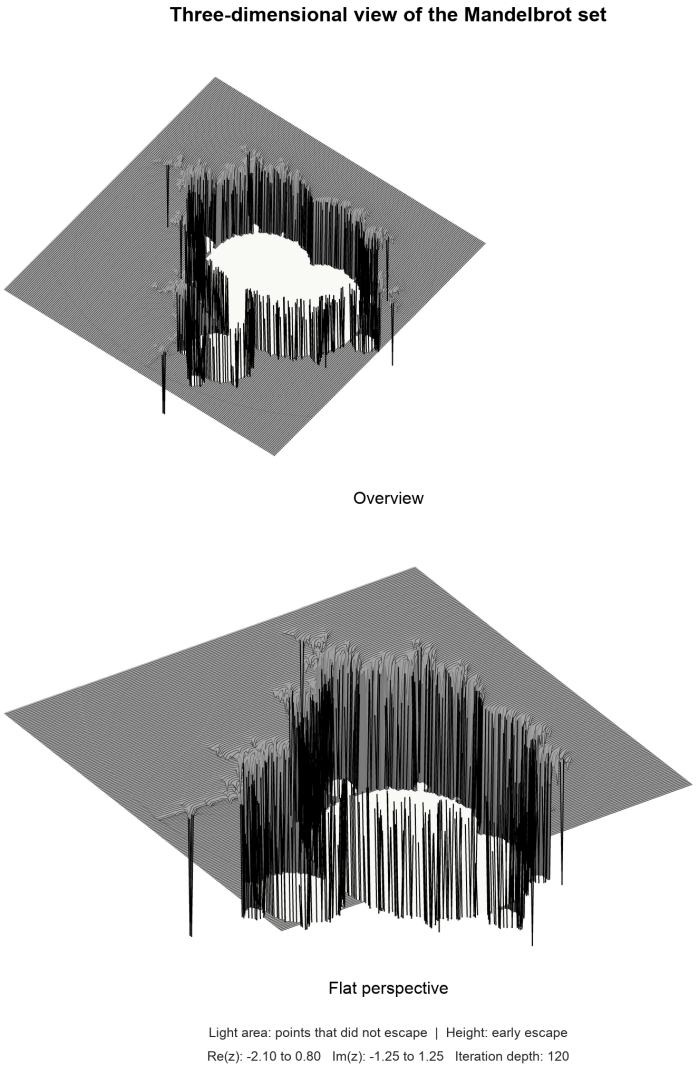
Plate 9: Koch curve and scaling (original page 39)



	Shape	Number N	Scaling s	Law
	Line segment	3	3	$3^1 = 3$
	Square	9	3	$3^2 = 9$
	Cube	27	3	$3^3 = 27$

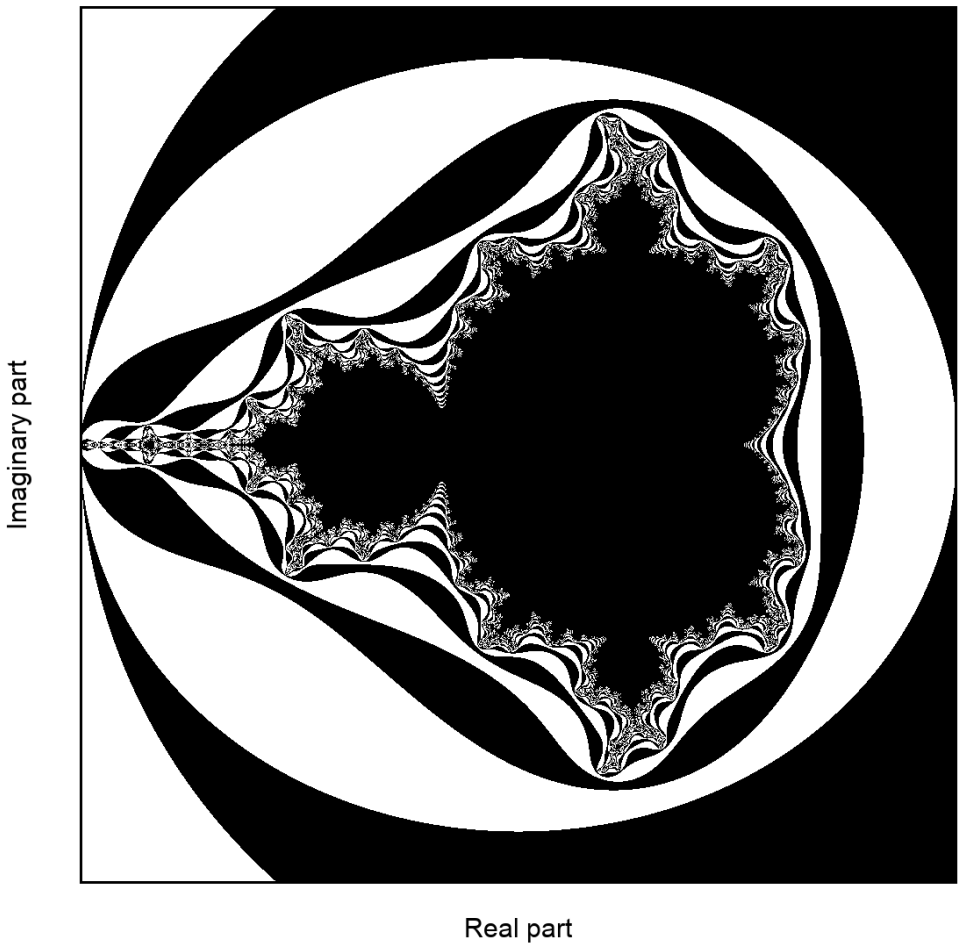
The iteration stages of the Koch curve and the table for line segment, square and cube illustrate the scaling law $N = s^D$. The curve and the table were redrawn from the original in 2026; the original's symbols a and d have been brought into line with the notation $N = s^D$ used in Section 4.2. The Koch curve is generated by a different rule from the Mandelbrot set. It serves here as a clear example on which self-similarity and a fractional dimension can be calculated directly.

Plate 10: Three-dimensional representation (original page 40)



The two perspectives were recomputed for $\mathrm{Re}(c) \in [-2.10, 0.80]$, $\mathrm{Im}(c) \in [-1.25, 1.25]$ and an iteration depth of 120; the original gives no coordinates for this plate, and the values were chosen in 2026. Points whose orbit does not escape within this limit form the light-coloured surface of the "Mandelbrot lake". For the remaining points, the height is chosen to be greater the earlier the orbit crosses the escape radius. The height is therefore not a third spatial coordinate of the Mandelbrot set but a visualisation of numerical data. Depending on the height function, the same computation can produce landscapes that look different; the planar Mandelbrot set remains unchanged.

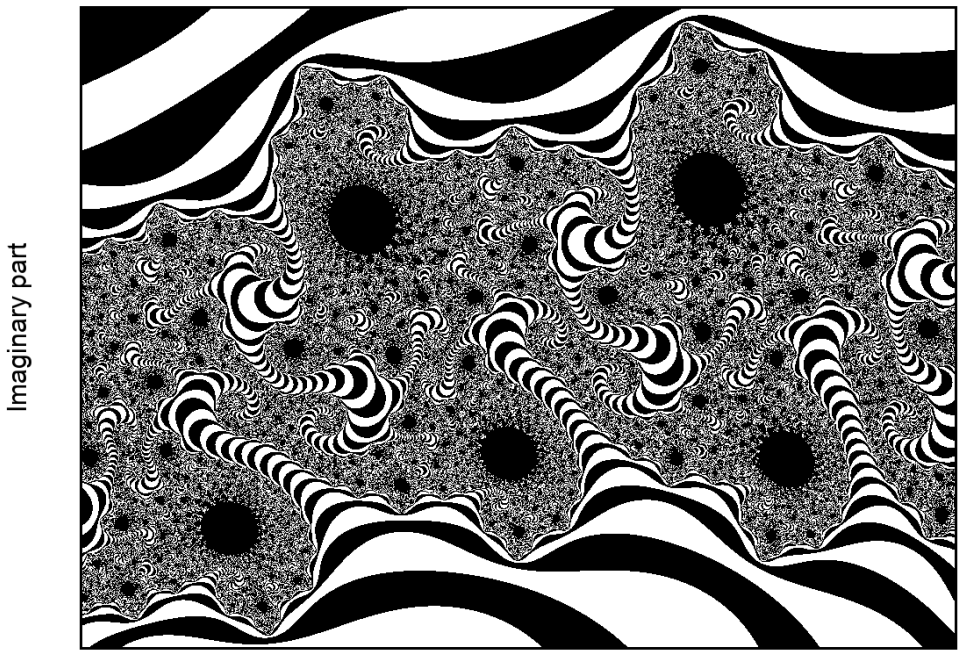
Plate 11: Differentiated graphical representation (original page 41)




```
   Real part:  Minimum -2.00000000   Maximum 1.00000000
Imaginary part:  Minimum -1.50000000   Maximum 1.50000000
                 Iteration depth: 25
```

The representation uses the differentiated colouring or shading explained in Section 3.4. Points with similar escape times are grouped into common bands. The outer rings and arcs thus show not only the boundary of the Mandelbrot set but also the behaviour of the escaping sequences in its neighbourhood.

Plate 12: Detail with high iteration depth (original page 42)




```
  Real part:  Minimum -0.74553800   Maximum -0.74505400
Imaginary part:  Minimum 0.11288100   Maximum 0.11323600
                   Iteration depth: 250
```

The detail was produced with an iteration depth of 250. Its width in the real part is only 0.000484. Despite this strong magnification, miniature copies and branching boundary structures appear once again. One can therefore speak of approximate self-similarity; an exact copy of the whole figure, however, does not occur.

10.2 Afterword

Are fractals merely mathematical games, or do they also prove their worth in practice? Fractal methods are in fact applied in various fields. Algorithms consisting of only a few instructions can generate artificial landscapes that are barely distinguishable from real ones. Such methods can be used in film production, for example, to supplement or replace elaborate models.

A further application lies in digital image transmission and image compression. Image information can be described compactly by suitable mathematical rules. This allows data to be transmitted faster, which matters particularly where storage space and transmission time are scarce or expensive.

The last chapters show that fractal geometry is an extensive field within mathematics. A Facharbeit can give only a general insight. The limits of school mathematical knowledge are quickly reached, since school mathematics is not sufficient for many of the details. The references list several works that deal with particular fractal structures and related topics.

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Note on sources The list follows the order and selection of the original scan. In 2026 the details were checked against library catalogues and publishers’ information and supplemented with subtitles and series. Entry [3] has been corrected: *The Science of Fractal Images* is a volume edited by Peitgen and Saupe and appeared in 1988, not 1986. The original gives no page numbers for the individual statements and figures used. Entries [1] to [9] form the 1990 list; from [10] onwards follow the sources used in 2026 for the further chapters 5 to 9.

10.4 Pascal listing

The printout of the program runs to eleven sheets: the first begins with `PROGRAM FRAKTAL;`, the last ends with `END.`. The following listing is a character-for-character transcription of this printout from 1990: comments, umlauts, spellings (including typing errors), brackets and indentation are reproduced as printed. Only lines that the printer had wrapped after 80 characters have been rejoined; the character hidden at the edge of the paper in the process was supplied from the context. The comments and on-screen texts in the listing are in German, as printed in 1990.

A complete compilation is nevertheless not possible, because the included files `GEMCONST.PAS`, `TRIXCONST.PAS`, `GEMTYPE.PAS`, `TRIXTYPE.PAS`, `GEMSUBS` and `TRIXSUBS` are not available. Routines such as `PAINT_RECT`, `GET_SCREEN` and `INIT_GEM` also come from them. The listing therefore documents the accessible source code, but without these contemporary libraries it is not a self-contained, compilable program.

Original Pascal source code from 1990

```
PROGRAM FRAKTAL;

 CONST

  {$I GEMCONST.PAS}         { Quelltext - Konstanten             }
  {$I TRIXCONST.PAS}        { Zusätzliche Quelltext - Konstanten }

 TYPE

  {$I GEMTYPE.PAS}          { Quelltext - Datentypen             }
  {$I TRIXTYPE.PAS}         { Zusätzliche Quelltext - Datentypen }

 VAR TASTE         : CHAR;           { einfacher Character                }
     I             : INTEGER;        { Wird im Zusammenhang mit Scan ver-
                                       wendet. Entspricht dem Zahlenwert
                                       der jeweiligen Taste                 }

     BILDSCHIRM    : SCREEN_TYPE;    { ST spezielles Bildschirm - Fomat    }
     BILD_SPEICHER : BOOLEAN;        { überprüft ob sich ein Bild im
                                       Speicher befindet. false entspricht
                                       einem leeren und true einem vollen
                                       Bildschirm-Speicher.                 }
     EN            : BOOLEAN;

  {$I GEMSUBS}              { Quelltext - Prozedur- und Funktionsköpfe }
  {$I TRIXSUBS}             { Weitere Prozedur- und Funktionsköpfe     }


 { Diese Routine fragt die Scancodes der Tastatur ab. Auf Grund dieser
   Routine ist es möglich auch spezielle Tasten (z.B. Funktionstasten)
   abzufragen. Dies ist mit dem herkömmlichen readln - Befehl nicht
   möglich.                                                          }

 FUNCTION BCONCIN(DEV : INTEGER) : LONG_INTEGER;BIOS(2);

 FUNCTION SCAN : INTEGER;

  BEGIN
   SCAN:=INT(BCONCIN (2) DIV (65535))
  END;

 { Prozedur zur Kursorsteuerung }

 PROCEDURE CURSOR(FUNKTION,RATE : INTEGER);XBIOS(21);
```

```
45
46
47  {-------------------------------------------------------------------------
48   FRACTALE AUSDRUCKEN
49   -------------------------------------------------------------------------}
50
51  PROCEDURE DRUCKEN;
52
53   CONST X_MAX = 639;
54         Y_MAX = 399;
55
56
57   VAR X,Y,I,J,CODE : INTEGER;
58       OK           : ARRAY[1..8] OF BOOLEAN;
59
60   BEGIN
61    IF (BILD_SPEICHER = FALSE) THEN
62     BEGIN
63      CUR_OFF;
64      CLEAR_SCREEN;
65      HIDE_MOUSE;
66      PAINT_STYLE(0);
67      PAINT_RECT(1,1,638,398);
68      PAINT_STYLE(1);
69      PAINT_RECT(4,4,632,392);
70      PAINT_STYLE(5);
71      PAINT_RECT(5,5,630,390);
72      PAINT_STYLE(1);
73      PAINT_RECT(232,189,194,34);
74      PAINT_STYLE(0);
75      PAINT_RECT(228,185,190,30);
76      GOTOXY(30,12);WRITELN('KEIN BILD IM SPEICHER');
77      I:=SCAN;
78     END
79    ELSE
80     BEGIN
81      PUT_SCREEN(BILDSCHIRM);
82      REWRITE(OUTPUT,'PRN:');    { Initialisierung des Druckers }
83      WRITELN(CHR(27),'U1');
84      WRITELN(CHR(27),'3',CHR(16));
85      X:=0;
86      Y:=0;
87      REPEAT
88       write(chr(27),'*',chr(6),chr((x_max+40+1) mod(256)),chr((x_max+40+1)div(256)));
89       for j:=1 to 40 do write(chr(0)); {linker Rand}
90       for x:=0 to x_max do
91        BEGIN
92         FOR I:=1 TO 8 DO OK[I]:=FALSE;
93         FOR I:=1 TO 8 DO
94          BEGIN
95           IF GET_PIXEL(X,Y+I-1)=BLACK THEN OK[I]:=TRUE;
96           IF I=8 THEN
97            BEGIN
98             CODE:=0;
99             IF OK[1]=TRUE THEN CODE:=CODE+128;
100            IF OK[2]=TRUE THEN CODE:=CODE+64;
101            IF OK[3]=TRUE THEN CODE:=CODE+32;
102            IF OK[4]=TRUE THEN CODE:=CODE+16;
103            IF OK[5]=TRUE THEN CODE:=CODE+8;
104            IF OK[6]=TRUE THEN CODE:=CODE+4;
105            IF OK[7]=TRUE THEN CODE:=CODE+2;
106            IF OK[8]=TRUE THEN CODE:=CODE+1;
107            WRITE(CHR(CODE));
108            IF X = X_MAX THEN FOR J:=1 TO (90*8-(X_MAX+1)-40) DO WRITE(CHR(0));
109           END;
110         END;
111       END;
```

```
      WRITELN(CHR(27),'J',CHR(8));
      Y:=Y+8;
    UNTIL Y>Y_MAX+8;
    REWRITE(OUTPUT,'CON:');
   END;
 END;

{-------------------------------------------------------------------------
 FRACTALE BERECHNEN
 -------------------------------------------------------------------------}

PROCEDURE BERECHNEN;

 VAR X_KOORDINATE,                  { x - Koordinate }
     Y_KOORDINATE,                  { y - Koordinate }
     RECHENTIEFE,                   { Rechentiefe    }
     ZAEHLER,                       { Zähler für Iterationen }
     X_Z,                           { Zähler für x - Koordinate }
     Y_Z           : INTEGER;       { Zähler für y - Koordinate }
     REAL_MIN,                      { Realteil - Minimum }
     REAL_MAX,                      { Realteil - Maximum }
     IMAG_MIN,                      { Imaginärteil - Minimum }
     IMAG_MAX,                      { Imaginärteil - Maximum }
     X_SCHRITT,                     { Schrittweite in x - Richtung }
     Y_SCHRITT,                     { schrittweite in y - Richtung }
     I_KONST,                       { Imaginärteil der Konstante }
     R_KONST,                       { Realteil der Konstante }
     REAL_ZAHL,                     { Neu errechneter Realteil }
     IMAG_ZAHL,                     { Neu errechneter Imaginärteil }
     REAL_Q,                        { Quadrierter Realteil }
     IMAG_Q       : REAL;           { Quadrierter Imaginäreil }
     STD,MIN,SEC  : INTEGER;        { Variabeln für Rechenzeit }


   {-------------------------------------------------------------------------
    GESTALTUNG DES HINTERGRUNDS UND DER EINGABEMASKE
    -------------------------------------------------------------------------}

   PROCEDURE HINTERGRUND;

    VAR A : INTEGER;

    BEGIN
     CUR_OFF;
     CLEAR_SCREEN;
     HIDE_MOUSE;
     PAINT_STYLE(0);
     PAINT_RECT(1,1,638,398);
     PAINT_STYLE(1);
     PAINT_RECT(4,4,632,392);
     PAINT_STYLE(5);
     PAINT_RECT(5,5,630,390);

     { Gestaltung des Eingabefensters }

     PAINT_STYLE(0);
     PAINT_RECT(113,50,413,65);
     PAINT_STYLE(1);
     PAINT_RECT(116,53,407,59);
     PAINT_STYLE(5);
     PAINT_RECT(117,54,405,57);
     PAINT_STYLE(1);
     PAINT_RECT(125,60,389,45);
     PAINT_STYLE(0);
     PAINT_RECT(123,58,389,45);
     TEXT_STYLE(0);
     TEXT_HEIGTH(13);
```

```
179       DRAW_STRING(235,85,'  Fractale berechnen  ');
180
181       { Gestaltung der Eingabebox }
182
183       PAINT_STYLE(0);
184       PAINT_RECT(113,120,413,246);
185       PAINT_STYLE(1);
186       PAINT_RECT(116,123,407,240);
187       PAINT_STYLE(5);
188       PAINT_RECT(117,124,405,238);
189       PAINT_STYLE(1);
190       PAINT_RECT(165,160,309,156);
191       PAINT_STYLE(0);
192       PAINT_RECT(163,158,309,156);
193       FOR A:=0 TO 3 DO
194        BEGIN
195         PAINT_STYLE(1);
196         PAINT_RECT(172,174+(A*32),143,25);
197         PAINT_STYLE(0);
198         PAINT_RECT(170,172+(A*32),143,25);
199         PAINT_STYLE(1);
200         PAINT_RECT(323,174+(A*32),140,25);
201         PAINT_STYLE(0);
202         PAINT_RECT(321,172+(A*32),140,25);
203        END;
204       GOTOXY(22,11);WRITELN('Realteil Minimum');
205       GOTOXY(22,13);WRITELN('Realteil Maximum');
206       GOTOXY(22,15);WRITELN('Imaginär Minimum');
207       GOTOXY(22,17);WRITELN('Imaginär Maximum');
208      END;
209
210
211  BEGIN
212
213   { Eingabe der Parameter und Fehlerüberprüfung }
214
215   REPEAT
216    HINTERGRUND;
217    CUR_ON;
218    CURSOR(3,0);
219    GOTOXY(42,11);READLN(REAL_MIN);
220    GOTOXY(42,13);READLN(REAL_MAX);
221    GOTOXY(42,15);READLN(IMAG_MIN);
222    GOTOXY(42,17);READLN(IMAG_MAX);
223    IF (REAL_MIN >= REAL_MAX) OR (IMAG_MIN >= IMAG_MAX) THEN
224     BEGIN
225      CUR_OFF;
226      PAINT_STYLE(1);
227      PAINT_RECT(165,160,309,156);
228      PAINT_STYLE(0);
229      PAINT_RECT(163,158,309,156);
230      DRAW_STRING(200,85,'E I N G A B E F E H L E R  !!!');
231      DRAW_STRING(210,200,'Achtung ! Das Minimum darf ');
232      DRAW_STRING(210,230,'nie größer als das Maximum ');
233      DRAW_STRING(210,260,'          sein !           ');
234      I:=SCAN;
235      CUR_ON;
236     END;
237   UNTIL (REAL_MAX > REAL_MIN) AND (IMAG_MAX > IMAG_MIN);
238
239   { Gestaltung des Hintergrunds }
240
241   CUR_OFF;
242   CLEAR_SCREEN;
243   HIDE_MOUSE;
244   PAINT_STYLE(0);
245   PAINT_RECT(1,1,638,398);
```

```
246  PAINT_STYLE(1);
247  PAINT_RECT(4,4,632,392);
248  PAINT_STYLE(5);
249  PAINT_RECT(5,5,630,390);
250
251  { Gestaltung der Eingabebox }
252
253  PAINT_STYLE(0);
254  PAINT_RECT(113,50,413,65);
255  PAINT_STYLE(1);
256  PAINT_RECT(116,53,407,59);
257  PAINT_STYLE(5);
258  PAINT_RECT(117,54,405,57);
259  PAINT_STYLE(1);
260  PAINT_RECT(125,60,389,45);
261  PAINT_STYLE(0);
262  PAINT_RECT(123,58,389,45);
263  TEXT_STYLE(0);
264  TEXT_HEIGTH(13);
265  DRAW_STRING(235,85,'  Fractale berechnen  ');
266  PAINT_STYLE(0);
267  PAINT_RECT(113,120,413,246);
268  PAINT_STYLE(1);
269  PAINT_RECT(116,123,407,240);
270  PAINT_STYLE(5);
271  PAINT_RECT(117,124,405,238);
272  PAINT_STYLE(1);
273  PAINT_RECT(165,160,309,156);
274  PAINT_STYLE(0);
275  PAINT_RECT(163,158,309,156);
276  CUR_ON;
277  GOTOXY(23,11);WRITELN('Bitte wählen Sie einen geeigneten');
278  GOTOXY(23,12);WRITELN('     Wert für die Anzahl der     ');
279  GOTOXY(23,13);WRITELN('          Iterationen !          ');
280  CUR_OFF;
281  TEXT_HEIGTH(4);
282  DRAW_STRING(175,230,'ACHTUNG !!!');
283  DRAW_STRING(175,238,'Wählen Sie für die Anzahl der Iterationen einen');
284  DRAW_STRING(175,246,'kleinen Wert, so ensteht ein grobes Bild ohne  ');
285  DRAW_STRING(175,254,'Feinstruktur. Bei größeren Werten sind Fein-    ');
286  DRAW_STRING(175,262,'strukturen zu erkennen, die aber lange Rechen- ');
287  DRAW_STRING(175,270,'zeiten beanspruchen.(=> Werte zwischen 100 und ');
288  DRAW_STRING(175,278,'350 günstig )                                  ');
289
290  CUR_ON;
291  GOTOXY(37,18);WRITELN('>');
292  REPEAT
293   GOTOXY(38,18);READLN(RECHENTIEFE);
294  UNTIL RECHENTIEFE>0;
295  CUR_OFF;
296  HIDE_MOUSE;
297  CLEAR_SCREEN;
298
299  { Rechenzeit auf Null setzen }
300
301  STD:=0;MIN:=0;SEC:=0;
302  SETTIME(STD,MIN,SEC);
303
304  {------------------------------------------------------------------------
305   DER APFELMÄNNCHEN-ALGORITHMUS
306    ------------------------------------------------------------------------}
307
308  X_KOORDINATE:=400;
309  Y_KOORDINATE:=400;
310
311  { Berechnung der x- und y-Schrittweite }
312
```

```
313   X_SCHRITT:=ABS(((REAL_MAX)-(REAL_MIN))/X_KOORDINATE);
314   Y_SCHRITT:=ABS(((IMAG_MAX)-(IMAG_MIN))/Y_KOORDINATE);
315
316   I_KONST:=IMAG_Min+Y_SCHRITT;
317   FOR Y_Z:=0 TO Y_KOORDINATE DO
318    BEGIN
319     R_KONST:=REAL_MIN+X_SCHRITT;
320     FOR X_Z:=0 TO X_KOORDINATE DO
321      BEGIN
322       REAL_ZAHL:=R_KONST;
323       IMAG_ZAHL:=I_KONST;
324       ZAEHLER:=RECHENTIEFE;
325       REPEAT
326        REAL_Q:=REAL_ZAHL*REAL_ZAHL;
327        IMAG_Q:=IMAG_ZAHL*IMAG_ZAHL;
328        IMAG_ZAHL:=2*REAL_ZAHL*IMAG_ZAHL+I_KONST;
329        REAL_ZAHL:=REAL_Q-IMAG_Q+R_KONST;
330        ZAEHLER:=ZAEHLER-1;
331       UNTIL (REAL_Q+IMAG_Q>4) OR (ZAEHLER=0);
332       IF ODD(ZAEHLER) OR (ZAEHLER=0)THEN PUT_PIXEL(X_Z,Y_Z,1);
333       R_KONST:=R_KONST+X_SCHRITT;
334      END;
335    I_KONST:=I_KONST+Y_SCHRITT;
336   END;
337
338   { Parameterbeschriftung }
339
340   GOTOXY(55,3 );WRITELN('Realteil:');
341   GOTOXY(55,4 );WRITELN('Minimum: ',REAL_MIN:8:8);
342   GOTOXY(55,5 );WRITELN('Maximum: ',REAL_MAX:8:8);
343   GOTOXY(55,7 );WRITELN('Imaginärteil:');
344   GOTOXY(55,8 );WRITELN('Minimum: ',IMAG_MIN:8:8);
345   GOTOXY(55,9 );WRITELN('Maximum: ',IMAG_MAX:8:8);
346   GOTOXY(55,11);WRITELN('Rechentiefe: ',RECHENTIEFE);
347   GETTIME(STD,MIN,SEC);
348   GOTOXY(55,13);WRITELN('Rechenzeit:  ',STD,':',MIN,':',SEC);
349   GET_SCREEN(BILDSCHIRM); { Aktueller Bildschirminhalt wird im
350                             Bildschirmspeicher abgelegt }
351   BILD_SPEICHER:=TRUE;
352   I:=SCAN;
353   SHOW_MOUSE;
354  END;
355
356  {--------------------------------------------------------------------------
357   AUSGABE DER AKTUELLEN PARAMETER
358   --------------------------------------------------------------------------}
359
360  PROCEDURE PARAMETER;
361
362   BEGIN
363    IF (BILD_SPEICHER = FALSE) THEN
364     BEGIN
365      CUR_OFF;
366      CLEAR_SCREEN;
367      HIDE_MOUSE;
368      PAINT_STYLE(0);
369      PAINT_RECT(1,1,638,398);
370      PAINT_STYLE(1);
371      PAINT_RECT(4,4,632,392);
372      PAINT_STYLE(5);
373      PAINT_RECT(5,5,630,390);
374      PAINT_STYLE(1);
375      PAINT_RECT(232,189,194,34);
376      PAINT_STYLE(0);
377      PAINT_RECT(228,185,190,30);
378      GOTOXY(30,12);WRITELN('KEIN BILD IM SPEICHER');
379     END
```

```
  ELSE
   PUT_SCREEN(BILDSCHIRM);
   I:=SCAN;
 END;

{-------------------------------------------------------------------------
 FRACTALE SPEICHERN UND LADEN
 -------------------------------------------------------------------------}

PROCEDURE IM_EXPORT;

 VAR FEHLER     : ERR_CODE;      { Fehler-Code beim Laden/Speichern }
     PFAD       : STRING;        { Pfad-Name der Datei              }
     NAME       : STRING;        { Name der Datei                   }
     D          : INTEGER;

BEGIN

 REPEAT

  { Gestaltung des Hintergrunds }

  CLEAR_SCREEN;
  PAINT_STYLE(0);
  PAINT_RECT(1,1,638,398);
  PAINT_STYLE(1);
  PAINT_RECT(4,4,632,392);
  PAINT_STYLE(5);
  PAINT_RECT(5,5,630,390);

  { Gestaltung der Eingabebox }

  PAINT_STYLE(0);
  PAINT_RECT(213,100,213,65);
  PAINT_STYLE(1);
  PAINT_RECT(216,103,207,59);
  PAINT_STYLE(5);
  PAINT_RECT(217,104,205,57);
  PAINT_STYLE(1);
  PAINT_RECT(225,110,189,45);
  PAINT_STYLE(0);
  PAINT_RECT(223,108,189,45);
  TEXT_STYLE(0);
  TEXT_HEIGTH(13);
  DRAW_STRING(235,125,'  Erstellte Bilder  ');
  DRAW_STRING(235,144,'Im- bzw. Exportieren');

  { Unterer Kasten in der Mitte }

  PAINT_STYLE(0);
  PAINT_RECT(213,170,213,150);
  PAINT_STYLE(1);
  PAINT_RECT(216,173,207,144);
  PAINT_STYLE(5);
  PAINT_RECT(217,174,205,142);
  FOR D:=0 TO 2 DO BEGIN
   PAINT_STYLE(0);
   PAINT_RECT(220,185+(D*42),40,39);
   PAINT_STYLE(1);
   PAINT_RECT(225,190+(D*42),32,31);
   PAINT_STYLE(0);
   PAINT_RECT(223,188+(D*42),32,31);
   PAINT_RECT(265,185+(D*42),154,39);
   PAINT_STYLE(1);
   PAINT_RECT(270,190+(D*42),146,31);
   PAINT_STYLE(0);
   PAINT_RECT(268,188+(D*42),146,31);
```

```
447   END;
448
449   { Beschriftung der Menüboxen }
450
451   TEXT_STYLE(0);
452   TEXT_HEIGTH(13);
453   DRAW_STRING(227,208,'F 6');
454   DRAW_STRING(227,250,'F 7');
455   DRAW_STRING(227,292,'ESC');
456   TEXT_HEIGTH(4);
457   DRAW_STRING(278,200,'   Erstellte Bilder   ');
458   DRAW_STRING(278,209,'importieren bzw. laden');
459   DRAW_STRING(278,238,'   Erstellte Bilder   ');
460   DRAW_STRING(278,246,'   exportieren bzw.   ');
461   DRAW_STRING(278,254,'      speichern       ');
462   DRAW_STRING(278,288,' Zurück ins Hauptmenü ');
463   I:=SCAN;
464   IF I=64 THEN
465    BEGIN
466     SHOW_MOUSE;
467     NAME:='';
468     PFAD:='A:\*.*';
469     IF GET_IN_FILE(PFAD,NAME) THEN
470      BEGIN
471       CLEAR_SCREEN;
472       FEHLER:=READ_SCREEN(1,NAME,BILDSCHIRM);
473       PUT_SCREEN(BILDSCHIRM);
474       BILD_SPEICHER:=TRUE;
475       I:=SCAN;
476      END;
477    END;
478   IF I=65 THEN
479    BEGIN
480     IF BILD_SPEICHER = FALSE THEN
481      BEGIN
482       CUR_OFF;
483       CLEAR_SCREEN;
484       HIDE_MOUSE;
485       PAINT_STYLE(0);
486       PAINT_RECT(1,1,638,398);
487       PAINT_STYLE(1);
488       PAINT_RECT(4,4,632,392);
489       PAINT_STYLE(5);
490       PAINT_RECT(5,5,630,390);
491       PAINT_STYLE(1);
492       PAINT_RECT(232,189,194,34);
493       PAINT_STYLE(0);
494       PAINT_RECT(228,185,190,30);
495       GOTOXY(30,12);WRITELN('KEIN BILD IM SPEICHER');
496       I:=SCAN;
497      END
498     ELSE
499      BEGIN
500       SHOW_MOUSE;
501       CLEAR_SCREEN;
502       PARAMETER;
503       NAME:='';
504       PFAD:='A:\*.*';
505       IF GET_IN_FILE(PFAD,NAME) THEN
506        BEGIN
507         FEHLER:=WRITE_SCREEN(1,NAME,BILDSCHIRM);
508        END;
509      END;
510    END;
511  UNTIL I=1;
512 END;
513
```

```
{------------------------------------------------------------------------
 INFORMATION
 ------------------------------------------------------------------------}

PROCEDURE INFO;

 BEGIN
  CLEAR_SCREEN;
  PAINT_STYLE(0);
  PAINT_RECT(1,1,638,398);
  PAINT_STYLE(1);
  PAINT_RECT(4,4,632,392);
  PAINT_STYLE(5);
  PAINT_RECT(5,5,630,390);
  PAINT_STYLE(0);
  PAINT_RECT(50,40,540,320);
  PAINT_STYLE(1);
  PAINT_RECT(60,50,520,300);
  PAINT_STYLE(0);
  PAINT_RECT(58,48,520,300);
  TEXT_HEIGTH(13);
  TEXT_STYLE(0);
  DRAW_STRING(70,70,'Information');
  DRAW_STRING(70,110,'Das vorliegende Computerprogramm ermöglicht die graphische');
  DRAW_STRING(70,130,'Darstellung der Mandelbrot-Menge. Hierzu müssen in F2 fünf');
  DRAW_STRING(70,150,'Eingabegrößen festgelegt werden. Bitte beachten Sie bei der');
  DRAW_STRING(70,170,'Eingabe,daß Sie keine Buchstaben eingeben. Verwenden Sie ');
  DRAW_STRING(70,190,'für Kommazahlen einen Punkt. Bei einer falschen Eingabe kann');
  DRAW_STRING(70,210,'durchaus vorkommen,daß das Programm abstürzt. Dieses schwer-');
  DRAW_STRING(70,230,'wiegende Problem konnte nicht gelöst werden, da es sich um');
  DRAW_STRING(70,250,'eine aufwendige Routine handelt ( Die Eingabe müßte als String');
  DRAW_STRING(70,270,'deklariert werden und auf jede Stelle untersucht werden ).');
  DRAW_STRING(70,290,'Dieses Programm wurde in Pascal erstellt. Pascal ist im Gegen-');
  DRAW_STRING(70,310,'satz zu Assembler relativ langsam. Dennoch wünsche ich dem ');
  DRAW_STRING(70,330,'Benutzer beim Erstellen von Fractale viel Spaß. (TASTE !)');
  I:=SCAN;
 END;

{------------------------------------------------------------------------
 HAUPTMENÜ
 ------------------------------------------------------------------------}
PROCEDURE HAUPTMENU;

 VAR A : INTEGER;

 BEGIN

  { Gestaltung des Hintegrunds }

  CLEAR_SCREEN;
  PAINT_STYLE(0);
  PAINT_RECT(1,1,638,398);
  PAINT_STYLE(1);
  PAINT_RECT(4,4,632,392);
  PAINT_STYLE(5);
  PAINT_RECT(5,5,630,390);

  { Fenster in der Mitte }

  PAINT_STYLE(0);
  PAINT_RECT(182,20,280,360);
  PAINT_RECT(184,22,276,356);
  PAINT_STYLE(4);
  PAINT_RECT(185,23,274,354);

  { Hintergrund der Menübox }

```

```
  PAINT_STYLE(0);
  PAINT_RECT(190,28,264,344);
  PAINT_RECT(192,30,260,340);
  PAINT_RECT(193,31,258,338);
  PAINT_STYLE(5);
  PAINT_RECT(194,32,256,336);

  { Beschriftung der Menübox }

  PAINT_STYLE(0);
  PAINT_RECT(199,40,245,55);
  PAINT_RECT(201,42,241,51);
  TEXT_STYLE(1);
  TEXT_HEIGTH(13);
  DRAW_STRING(250,65,'MANDELBROT - MENGEN');
  TEXT_HEIGTH(4);
  DRAW_STRING(220,78,'Das Computerprogramm zur Facharbeit');
  DRAW_STRING(220,88,'         von Urban Bettag          ');
  FOR A:=1 TO 6 DO
   BEGIN
    PAINT_STYLE(0);
    PAINT_RECT(199,79+(A*32),50,29);PAINT_RECT(279,79+(A*32),146,29);
    PAINT_STYLE(1);
    PAINT_RECT(204,84+(A*32),42,21);PAINT_RECT(284,84+(A*32),138,21);
    PAINT_STYLE(0);
    PAINT_RECT(202,82+(A*32),42,21);PAINT_RECT(282,82+(A*32),138,21);
   END;
  TEXT_STYLE(0);
  TEXT_HEIGTH(13);
  DRAW_STRING(212,130,'F 1');TEXT_HEIGTH(4);DRAW_STRING(288,122,'   Erstelle Bilder  ');
                                            DRAW_STRING(292,130,'Im- bzw. Exportieren');
  TEXT_HEIGTH(13);
  DRAW_STRING(212,162,'F 2');TEXT_HEIGTH(4);DRAW_STRING(292,158,' Fractale berechnen ');
  TEXT_HEIGTH(13);
  DRAW_STRING(212,194,'F 3');TEXT_HEIGTH(4);DRAW_STRING(292,186,' Aktuelle Parameter ');
                                            DRAW_STRING(292,193,'     einblenden     ');
  TEXT_HEIGTH(13);
  DRAW_STRING(212,226,'F 4');TEXT_HEIGTH(4);DRAW_STRING(292,222,' Fractal ausdrucken ');
  TEXT_HEIGTH(13);
  DRAW_STRING(212,258,'F 5');TEXT_HEIGTH(4);DRAW_STRING(274,294,'  Information für   ');
                                            DRAW_STRING(292,258,'     Benutzer       ');
  TEXT_HEIGTH(13);
  DRAW_STRING(212,290,'F10');TEXT_HEIGTH(4);DRAW_STRING(292,286,'  Programm beenden  ');
  TEXT_HEIGTH(13);
 END;

{------------------------------------------------------------------------
 BEENDEN
 ------------------------------------------------------------------------}

 PROCEDURE BEENDEN;

 BEGIN
  CLEAR_SCREEN;
  PAINT_STYLE(0);
  PAINT_RECT(1,1,638,398);
  PAINT_STYLE(1);
  PAINT_RECT(4,4,632,392);
  PAINT_STYLE(5);
  PAINT_RECT(5,5,630,390);
  PAINT_STYLE(1);
  PAINT_RECT(210,180,220,40);
  PAINT_STYLE(0);
  PAINT_RECT(208,178,220,40);
  DRAW_STRING(220,200,'Sind Sie sicher (J/N) ?');
  I:=SCAN;
  IF I=36 THEN EN:=FALSE;
```

```
  IF I=49 THEN EN:=TRUE;
 END;

{------------------------------------------------------------------------
 HAUPTPROGRAMM
 ------------------------------------------------------------------------}

BEGIN

 { GEM - Routinen werden aufgerufen und instaliert

   Init_Gem
   --------

   Die Funktion führt alle notwendigen Operationen durch, um GEM mitzuteilen
   daß nun eine Applikation an den Segnungen des Betriebssystems teilhaben
   möchte. Tritt ein Fehler auf so liefert die Funktion Init_gem der Wert -1
   Eine erfolgreiche Operation liefert immer einen positiven Wert.

   Exit_Gem
   --------

   Beendet ein GEM - Programm (Graphic Environment Manager). Die Prozedur
   teil GEM mit, daß ab nun keine weiteren GEM-Aufrufe getätigt werden.
   Bei einem fehlerhaften Aufruf von Init_Gem sollte man diese Prozedur nicht
   aufrufen. }

 IF INIT_GEM >=0 THEN
  BEGIN
   BILD_SPEICHER:=FALSE;
   EN:=TRUE;
   REPEAT
    HAUPTMENU;
    I:=SCAN;
    IF I=59 THEN IM_EXPORT;
    IF I=60 THEN BERECHNEN;
    IF I=61 THEN PARAMETER;
    IF I=62 THEN DRUCKEN;
    IF I=63 THEN INFO;
    IF I=68 THEN BEENDEN;
    SHOW_MOUSE;
   UNTIL EN=FALSE;
   EXIT_GEM;
  END;
END.
```

Notes on the program

- **Historical findings:** The program was evidently written for an Atari ST computer with the GEM interface and BIOS/XBIOS calls. Identifiers such as `TEXT_HEIGTH` have not been modernised, because they may come from the missing libraries.
- **Scan code:** The return value of `BCONCIN` contains the scan code in the upper word. This would call for a division by $65536 = 2^{16}$ or a shift by 16 bits. The printout reads `DIV (65535)`; this is very probably an error in the original program and has been deliberately retained.
- **Menu label:** In the main menu, the label “Information für” is at $y = 294$, and the corresponding second line “Benutzer” at $y = 258$. The two words therefore appear separately on the screen; this too is an error in the original and is preserved.

- **Grid bounds:** The loops each run from 0 up to and including 400, that is, over 401 values. At the same time, the real and imaginary parts are increased by one step before the first pixel. As a result, the stated minima are skipped and values above the maxima are reached at the end. In addition, the y-coordinate 400 may lie outside the intended grid for a screen area from 0 to 399.
- **Iteration:** For each parameter c, the computation in the program starts with $z = c$, not with $z_0 = 0$. For the pure escape test this amounts to a count shifted by one step; for the black-and-white colouring used, however, it changes the parity of the iteration count. Moreover, the termination sum `REAL_Q+IMAG_Q` is formed from the squares before the update just computed.
- **Representation:** Exterior points are drawn black or white according to the parity of the remaining counter. The stripes therefore show escape-time classes. They are not in themselves proof that a point belongs to the Mandelbrot set.
- **Present-day assessment:** With the missing GEM/TRIX files and a suitable historical Pascal system, the syntax could be checked more precisely. Without these components, any supposedly runnable reconstruction would be speculative; for this reason only the program text actually available has been included here.

10.5 Python post-processing from 2026

The following Python programs are not part of the original Facharbeit of 1990. They were written in 2026 for the digital revision. The first program computes the new black-and-white, colour and detail images of the Mandelbrot set. The second program produces the three-dimensional escape-time representation. Both programs use the mathematical iteration $z_{n+1} = z_n^2 + c$ with the starting value $z_0 = 0$.

In each case only the computational core is printed; the line numbers correspond to the complete file. The remaining parts of the programs merely produce labels and the image layout. The complete scripts `generate_fractals.py`, `generate_3d_fractal.py`, `generate_diagrams.py` and `generate_ifs.py` are not printed.

Escape times for the two-dimensional images

`generate_fractals.py`: function `escape_times`

```
40 def escape_times(bounds, iterations, width=1000):
41     """Berechne die erste Fluchtiteration fuer z_0=0, z_(n+1)=z_n^2+c."""
42     xmin, xmax, ymin, ymax = bounds
43     height = max(2, round(width * (ymax - ymin) / (xmax - xmin)))
44     x = np.linspace(xmin, xmax, width, dtype=np.float64)
45     y = np.linspace(ymin, ymax, height, dtype=np.float64)
46     c = x[np.newaxis, :] + 1j * y[:, np.newaxis]
47     z = np.zeros_like(c)
48     active = np.ones(c.shape, dtype=bool)
49     escaped = np.full(c.shape, iterations, dtype=np.uint16)
50 
51     for n in range(1, iterations + 1):
52         z[active] = z[active] * z[active] + c[active]
53         newly = active & (z.real * z.real + z.imag * z.imag > 4.0)
54         escaped[newly] = n
55         active[newly] = False
56         if not active.any():
57             break
58     return escaped
```

For each pixel c, the sequence is iterated from $z_0 = 0$ until $|z_n|^2 > 4$ holds or the iteration limit is reached. What is stored is the index of the first crossing, that is, the escape time.

Height field for the three-dimensional representation

`generate_3d_fractal.py`: function `height_field`

```
44 def height_field(nx=230, ny=190):
45     xmin, xmax, ymin, ymax = BOUNDS
46     xs = np.linspace(xmin, xmax, nx)
47     ys = np.linspace(ymin, ymax, ny)
48     c = xs[np.newaxis, :] + 1j * ys[:, np.newaxis]
49     z = np.zeros_like(c)
50     active = np.ones(c.shape, dtype=bool)
51     smooth = np.full(c.shape, float(MAX_ITER), dtype=np.float64)
52 
53     for iteration in range(1, MAX_ITER + 1):
54         z[active] = z[active] * z[active] + c[active]
55         radius2 = z.real * z.real + z.imag * z.imag
56         escaped = active & (radius2 > 4.0)
```

```
        if escaped.any():
            modulus = np.sqrt(radius2[escaped])
            smooth[escaped] = (
                iteration + 1.0 - np.log(np.log(modulus)) / np.log(2.0)
            )
        active[escaped] = False
        if not active.any():
            break

    # Schnelles Entweichen ergibt grosse Hoehen. Nicht entwichene Punkte sind
    # der ebene See auf Hoehe null. Die Wurzel macht die Kueste besser sichtbar.
    height = np.zeros_like(smooth)
    outside = ~active
    speed = np.clip((MAX_ITER - smooth[outside]) / MAX_ITER, 0.0, 1.0)
    height[outside] = 0.20 + 1.85 * np.sqrt(speed)
    return xs, ys, height, active
```

Here the escape time is smoothed and converted into a height: the faster a point escapes, the higher it lies. Points that have not escaped form the flat surface of the “Mandelbrot lake”.

Change log

The table lists the most important corrections of content on original pages 20 to 24. It is not exhaustive.

Page	Original problem	Correction	Effect
20	Starting value and index shifted	$z_0 = 0$, $z_1 = k$, $z_2 = k^2 + k$	Recursion and program count agree
21	Parameter k drops out	Case distinction explicitly restricted to $k = 0$	The existing section is kept as a special case
22	$\mathcal{M}$ defined via convergence to zero	Boundedness of the critical orbit	Correct definition of the set
22–23	Finite test taken as proof of membership	Non-escape up to N as an approximation	The algorithmic statement becomes precise
23	New components not written back	Update a and b after every step	The pseudocode actually performs an iteration
24	Parameter plane and starting-value plane conflated	Separate the Mandelbrot and Julia planes	The relationship between the two sets is described correctly

Page numbers refer to the 1990 original.

Source basis

The basis is the original scans `IMG_1917` to `IMG_1960`, covering printed pages 1 to 44. File number `IMG_1950` is missing; according to the author, the corresponding gap is closed by the colour plate `IMG_1960`, photographed later. For the partial check of the program, the visible program core in `IMG_1965` was also used. The historical corrections use mathematical foundations that were already available in 1990; later findings are explicitly marked as such.